SOURCE CODE DOCUMENTATION

Campus360

Educational Institution Management Information System (IMS) — Proof of Concept

🏛️ IEEE Research Paper Specifications Implemented

- **IEEE 10058674**: "*Institution Management System: Student Module*" — Record consolidation, dossier modal, promotion engine, placement & internship directorate.
- **IEEE 10516390**: "*Educational Institution Management Information System*" — Centralized multi-role web platform (Admin, Faculty, Student) with institutional broadcast engine.
- **IEEE 7384005**: "*College Student Management System Design Using Computer Aided System*" — 4-Tier Computer-Aided Architecture & Relational Entity-Relationship Model.

GITHUB REPOSITORY
HaydenVictor420/Campus360

GIT BRANCH
muhil

LOCAL SERVER URL
http://localhost:3000

TOTAL CODEBASE VOLUME
11 files • 4,192 total lines

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Comprehensive listing of all project source code modules, schemas, and configurations.

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#1 MARKDOWN **README.md** 134 lines • 6.7 KB

Project Overview, IEEE Research Mapping & System Architecture

```
# Campus360 — Educational Institution Management Information System (IMS)

[![Local Preview](https://img.shields.io/badge/Local_App-http%3A%2F%2Flocalhost%3A3000-brightgreen?style=for-the-badge&logo=googlechrome)](http://localhost:3000)
[![Live Demo](https://img.shields.io/badge/GitHub_Pages-Live_Demo-blue?style=for-the-badge&logo=github)](https://haydenvictor420.github.io/Campus360/)
[![Branch](https://img.shields.io/badge/Branch-Muhil-orange?style=for-the-badge&logo=git)](https://github.com/HaydenVictor420/Campus360/tree/Muhil)
[![Source Code PDF](https://img.shields.io/badge/Documentation-Source_Code_PDF-red?style=for-the-badge&logo=adobeacrobatreader)](./Campus360_Source_Code.pdf)

An evision project implementing an enterprise-grade, interactive Proof of Concept based on architectural designs, student module specifications, and entity-relationship models published across three IEEE research papers:

1. **IEEE 10058674**: *"Institution Management System: Student Module"*
2. **IEEE 10516390**: *"Educational Institution Management Information System"*
3. **IEEE 7384005**: *"College Student Management System Design Using Computer Aided System"*

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## 🌐 Quick Links & Access

### 🖥️ 1. Local Application (Default)
When the local server is running, access the full interactive system at:
👉 **[http://localhost:3000](http://localhost:3000)**

> [!NOTE]
> **If `http://localhost:3000` says *"Site can't be reached"***:
> The local Node server needs to be started on your computer:
> - **Option A (One-Click)**: Double-click **`start.bat`** in this project folder.
> - **Option B (Terminal)**: Run `npm start` (or `node server.js`) in your terminal.
>
> The server will boot instantly and automatically launch `http://localhost:3000` in your default browser.

### 🚀 2. Live Online Demo (GitHub Pages)
Explore Campus360 directly in your browser without needing to run any local server:
👉 **[https://haydenvictor420.github.io/Campus360/](https://haydenvictor420.github.io/Campus360/)**

### 📄 3. Complete Source Code PDF
Download or view the complete compiled codebase documentation with line numbers, syntax highlighting, and table of contents across all 9 modules (~3,700 lines):
👉 **[Download Campus360_Source_Code.pdf](./Campus360_Source_Code.pdf)**

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## 🏛️ IEEE Research Paper Architectural Mapping

| IEEE Publication | Core Contribution in Paper | Realization in this PoC |
| :--- | :--- | :--- |
| **IEEE 10058674** | **Student Module & Record Consolidation**: Eliminates record redundancy. Integrates admission details, transfer details, promotion history, course evaluations, corporate placement training, events attended, and student research works/publications. | Full **Student Dossier Modal**, **Promotion Progress Engine**, **Placement & Internship Directorate**, and **Research & Capstone Showcase** with real-time editing and state persistence. |
| **IEEE 10516390** | **Central Management Information System (MIS)**: Built using web technologies (PHP/JS/CSS). Centralizes multi-user data (administrators, instructors, students) to reduce administrative overhead and improve communication. | **Multi-Role Portal Switcher** (`Admin`, `Faculty`, `Student`), **Institutional Notice Board & Broadcast Hub**, and interactive KPI metrics for institutional governance. |
```

```
46  | **IEEE 7384005** | **4-Tier Computer-Aided System & ER Model**: Hierarchical separation into (1) Web Display Layer, (2) Business Logic Layer, (3) Data Access Layer, (4) Database Layer. Relational ER model connecting Students, Teachers, Departments, Majors, Classes, Courses, and Comprehensive Scores. | Strict modular 4-tier JavaScript implementation (`app.js`), interactive **4-Tier Architecture Viewer**, and a visual **ER Schema Explorer** with PK/FK definitions. |
47
48  ---
49
50  ## 🏗️ 4-Tier System Architecture (IEEE 7384005)
51
52  ```mermaid
53  graph TD
54      subgraph Tier 1: Web Display Layer
55          A1[Admin Management Portal]
56          A2[Faculty Grade & Supervision Portal]
57          A3[Student Self-Service Portal]
58      end
59
60      subgraph Tier 2: Business Logic Layer
61          B1[Comprehensive GPA Calculator]
62          B2[Semester Promotion Engine]
63          B3[Placement Eligibility Auditor]
64          B4[Role-Based Access Controller]
65      end
66
67      subgraph Tier 3: Data Access Layer (DAL)
68          C1[Student Repository]
69          C2[Schema Relational Mapper]
70          C3[JSON Export / Restore Engine]
71      end
72
73      subgraph Tier 4: Database Layer
74          D1[(Central Normalized Schema)]
75          D2[(Audit & Promotion Logs)]
76          D3[(Notice & Communication DB)]
77      end
78
79      Tier 1 --> Tier 2
80      Tier 2 --> Tier 3
81      Tier 3 --> Tier 4
82  ```
83
84  ---
85
86  ## 🧩 Comprehensive Entity-Relationship (ER) Schema
87
88  The data model follows the ER diagram specified in **IEEE 7384005**:
89
90  - `STUDENT_INFO`: `(PK: id, rollNo, name, email, FK: deptId, FK: majorId, FK: classId, cgpa, totalCreditsEarned, academicStatus, promotionStatus)`
91  - `TEACHER_INFO`: `(PK: id, name, department, title, email)`
92  - `DEPARTMENT_INFO`: `(PK: id, code, name, head)`
93  - `MAJOR_INFO`: `(PK: id, FK: deptId, name, degree)`
94  - `CLASS_INFO`: `(PK: id, name, currentSemester, FK: counselorId)`
95  - `COURSE_INFO`: `(PK: id, code, name, credits, type, dept, FK: instructorId)`
96  - `COMPREHENSIVE_SCORE`: `(Internal [30] + Assignments [20] + Final Exam [50] = Total Score [100] ➔ Grade Point & Letter Grade)`
97  - `PLACEMENT_INFO`: `(company, role, ctc, offerDate, status)`
98  - `RESEARCH_AND_PROJECTS`: `(title, type, venue, guide, status)`
99
100 ---
101
102 ## ⚡ Key Features
103
104 1. **Role-Based Simulation**:
```

```
    - **Administrator**: Full CRUD on student records, instant semester promotion, and campus-wide notice p
ublishing.
    - **Faculty**: Grade entry interface with automatic weighted GPA calculation.
    - **Student**: View enrolled modules, comprehensive breakdown, internships, and transcript history.
2. **Dynamic Search & Multi-Predicate Filtering**:
    - Instant live search across names, roll numbers, and emails.
    - Filter by academic department and placement/academic status.
3. **Comprehensive Score Engine**:
    - Calculates letter grades (`O`, `A+`, `A`, `B+`, `B`, `C`, `F`) and GPAs dynamically.
4. **Placement Directorate & Eligibility Audit**:
    - Automates candidate shortlisting based on CGPA thresholds and zero-backlog checks.
5. **Zero-Dependency Architecture**:
    - Pure HTML5, modern vanilla CSS, and vanilla JavaScript.
    - Persistent `localStorage` database with one-click JSON export and reset to seed data.

---

## 🚀 Running the PoC Locally

### Option A: One-Click Windows Script
Double-click **`start.bat`**.

### Option B: Built-in Zero-Dependency Node.js Server
```bash
node server.js
```
Then navigate to: **`http://localhost:3000`**

Option C: Direct Browser Launch
Open `index.html` directly in any modern browser (Chrome, Edge, Firefox, Safari).
```

#2 JSON package.json

21 lines • 0.6 KB

Project Metadata, Dependencies & NPM Scripts

```
{
  "name": "ims-student-poc",
  "version": "1.0.0",
  "description": "Educational Institution Management Information System (IMS) PoC based on IEEE 10058674, IEEE 10516390, and IEEE 7384005",
  "main": "server.js",
  "scripts": {
    "start": "node server.js",
    "dev": "node server.js",
    "app": "node app.js",
    "build:pdf": "node build-code-pdf.js"
  },
  "keywords": [
    "ieee",
    "institution-management",
    "student-information-system",
    "4-tier-architecture"
  ],
  "author": "Antigravity Engineering",
  "license": "ISC"
}

```

#3 JAVASCRIPT **server.js** 119 lines • 3.4 KB

Zero-Dependency HTTP Server & Port Failover

```
/**
 * Lightweight Zero-Dependency HTTP Server for IMS PoC
 * Serves static web assets and handles automatic port failover & browser launch
 */

const http = require('http');
const fs = require('fs');
const path = require('path');
const { exec } = require('child_process');

const DEFAULT_PORT = parseInt(process.env.PORT, 10) || 3000;
const PUBLIC_DIR = __dirname;

const MIME_TYPES = {
  '.html': 'text/html',
  '.css': 'text/css',
  '.js': 'text/javascript',
  '.json': 'application/json',
  '.png': 'image/png',
  '.jpg': 'image/jpeg',
  '.svg': 'image/svg+xml',
  '.ico': 'image/x-icon'
};

function createServerInstance() {
  return http.createServer((req, res) => {
    let reqUrl = req.url.split('?')[0];
    if (reqUrl === '/') reqUrl = '/index.html';

    const filePath = path.join(PUBLIC_DIR, reqUrl);

    // Prevent directory traversal
    if (!filePath.startsWith(PUBLIC_DIR)) {
      res.writeHead(403, { 'Content-Type': 'text/plain' });
      res.end('Access Denied');
      return;
    }

    fs.stat(filePath, (err, stats) => {
      if (err || !stats.isFile()) {
        res.writeHead(404, { 'Content-Type': 'text/plain' });
        res.end('404 Not Found');
        return;
      }

      const ext = path.extname(filePath).toLowerCase();
      const contentType = MIME_TYPES[ext] || 'application/octet-stream';

      res.writeHead(200, {
        'Content-Type': contentType,
        'Cache-Control': 'no-cache, no-store, must-revalidate'
      });

      const readStream = fs.createReadStream(filePath);
      readStream.pipe(res);
    });
  });
}
```

```
59
60  function openBrowserUrl(url) {
61    try {
62      const startCmd = process.platform === 'win32' ? `start "" "${url}"` :
63                       process.platform === 'darwin' ? `open "${url}"` :
64                       `xdg-open "${url}"`;
65      exec(startCmd, (err) => {
66        if (err) console.log(`Note: Please open ${url} in your browser.`);
67      });
68    } catch (e) {
69      // Silent fallback
70    }
71  }
72
73  function startServer(options = {}) {
74    const openBrowser = options.openBrowser !== false;
75    let currentPort = options.port || DEFAULT_PORT;
76    const maxPort = currentPort + 10;
77
78    function tryListen(port) {
79      const srv = createServerInstance();
80
81      srv.once('error', (err) => {
82        if (err.code === 'EADDRINUSE') {
83          console.warn(`Port ${port} is currently busy, attempting port ${port + 1}...`);
84          if (port < maxPort) {
85            tryListen(port + 1);
86          } else {
87            console.error(`Unable to bind to any port between ${DEFAULT_PORT} and ${maxPort}.`);
88          }
89        } else {
90          console.error('Server error:', err);
91        }
92      });
93
94      srv.listen(port, () => {
95        const url = `http://localhost:${port}`;
96        console.log('============================================================');
97        console.log(' Educational Institution Management System (IMS) PoC');
98        console.log(` Server is running live at: ${url}`);
99        console.log('============================================================');
100
101       if (openBrowser) {
102         console.log(`Opening ${url} in your default browser...`);
103         openBrowserUrl(url);
104       }
105     });
106
107     return srv;
108   }
109
110   return tryListen(currentPort);
111 }
112
113 // If run directly via "node server.js"
114 if (require.main === module) {
115   startServer({ openBrowser: true });
116 }
117
118 module.exports = { startServer };
119
```

#4 HTML **index.html** 250 lines • 10.7 KB

Web Display Layer (Tier 1) - Multi-Role Dashboard HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Institution Management System (IMS) - PoC | IEEE 10058674, 10516390, 7384005</title>
  <meta name="description" content="Proof of Concept for Educational Institution Management Information System and Student Module based on IEEE 10058674, IEEE 10516390, and IEEE 7384005.">

  <!-- Modern Typography from Google Fonts -->
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=JetBrains+Mono:wght@400;500;700&family=Plus+Jakarta+Sans:wght@400;500;600;700;800&display=swap" rel="stylesheet">

  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <!-- Top Application Navigation Header -->
  <header class="app-header">
    <div class="brand-section">
      <div class="brand-icon">
        <svg width="22" height="22" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round">
          <path d="M22 10v6M2 10l10-5 10 5-10 5z"></path>
          <path d="M6 12v5c3 3 9 3 12 0v-5"></path>
        </svg>
      </div>
      <div class="brand-text">
        <h1>Institution Management System (IMS)</h1>
        <div class="subtitle">
          <span>Student Module & Computer-Aided Architecture</span>
          <span class="badge-tag">IEEE 10058674</span>
          <span class="badge-tag">IEEE 10516390</span>
          <span class="badge-tag">IEEE 7384005</span>
        </div>
      </div>
    </div>

    <!-- Role Switcher & Action Bar -->
    <div class="header-actions">
      <div class="role-switcher-container">
        <span class="role-label">Active Role:</span>
        <button class="role-btn active" data-role="admin">
          <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
            <path d="M16 21v-2a4 4 0 0 0-4-4H5a4 4 0 0 0-4 4v2"></path>
            <circle cx="8.5" cy="7" r="4"></circle>
            <polyline points="17 11 19 13 23 9"></polyline>
          </svg>
          Admin
        </button>
        <button class="role-btn" data-role="faculty">
          <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
            <path d="M2 3h6a4 4 0 0 1 4 4v14a3 3 0 0 0-3-3H2z"></path>
            <path d="M22 3h-6a4 4 0 0 0-4 4v14a3 3 0 0 1 3-3h7z"></path>
```

```
            </svg>
            Faculty
          </button>
          <button class="role-btn" data-role="student">
            <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
              <path d="M20 21v-2a4 4 0 0 0-4-4H8a4 4 0 0 0-4 4v2"></path>
              <circle cx="12" cy="7" r="4"></circle>
            </svg>
            Student
          </button>
        </div>

        <button class="btn-quick" id="btn-export-data" title="Export current database as JSON">
          <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
            <path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"></path>
            <polyline points="7 10 12 15 17 10"></polyline>
            <line x1="12" y1="15" x2="12" y2="3"></line>
          </svg>
          Export
        </button>

        <button class="btn-quick" id="btn-reset-data" title="Reset to original IEEE dataset">
          <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
            <path d="M2.5 2v6h6M21.5 22v-6h-6"></path>
            <path d="M22 11.5A10 10 0 0 0 3.2 7.2M2 12.5a10 10 0 0 0 18.8 4.2"></path>
          </svg>
          Reset
        </button>

        <button class="btn-quick btn-primary" id="btn-new-student">
          <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2.
5">
            <line x1="12" y1="5" x2="12" y2="19"></line>
            <line x1="5" y1="12" x2="19" y2="12"></line>
          </svg>
          Enroll Student
        </button>
      </div>
    </header>

    <!-- Main Container -->
    <main class="app-container">

      <!-- Secondary Navigation / Module Tabs -->
      <nav class="tabs-navigation">
        <button class="tab-nav-btn active" data-tab="directory">
          <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
            <path d="M17 21v-2a4 4 0 0 0-4-4H5a4 4 0 0 0-4 4v2"></path>
            <circle cx="9" cy="7" r="4"></circle>
            <path d="M23 21v-2a4 4 0 0 0-3-3.87"></path>
            <path d="M16 3.13a4 4 0 0 1 0 7.75"></path>
          </svg>
          Student Module (Master Records)
        </button>

        <button class="tab-nav-btn" data-tab="academics">
          <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
            <path d="M22 12h-4l-3 9L9 3l-3 9H2"></path>
          </svg>
          Comprehensive Scores & Grades
```

```
        </button>

        <button class="tab-nav-btn" data-tab="placements">
          <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
            <rect x="2" y="7" width="20" height="14" rx="2" ry="2"></rect>
            <path d="M16 21V5a2 2 0 0 0-2-2h-4a2 2 0 0 0-2 2v16"></path>
          </svg>
          Placement & Internships
        </button>

        <button class="tab-nav-btn" data-tab="research">
          <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
            <polygon points="12 2 15.09 8.26 22 9.27 17 14.14 18.18 21.02 12 17.77 5.82 21.02 7 14.14 2 9.27
8.91 8.26 12 2"></polygon>
          </svg>
          Research Papers & Capstones
        </button>

        <button class="tab-nav-btn" data-tab="architecture">
          <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
            <polygon points="12 2 2 7 12 12 22 7 12 2"></polygon>
            <polyline points="2 17 12 22 22 17"></polyline>
            <polyline points="2 12 12 17 22 12"></polyline>
          </svg>
          4-Tier Architecture & ER Schema
        </button>

        <button class="tab-nav-btn" data-tab="communications">
          <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
            <path d="M21 15a2 2 0 0 1-2 2H7l-4 4V5a2 2 0 0 1 2-2h14a2 2 0 0 1 2 2z"></path>
          </svg>
          Campus Notice Hub
        </button>
      </nav>

      <!-- Key Performance Metrics Row -->
      <div class="hero-stats-grid">
        <div class="stat-card" style="--accent-color: var(--primary); --accent-light: var(--primary-ligh
t);">
          <div class="stat-header">
            <span class="stat-title">Enrolled Students</span>
            <div class="stat-icon">
              <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
                <path d="M17 21v-2a4 4 0 0 0-4-4H5a4 4 0 0 0-4 4v2"></path>
                <circle cx="9" cy="7" r="4"></circle>
              </svg>
            </div>
          </div>
          <div class="stat-value" id="metric-total-students">5</div>
          <div class="stat-footer">
            <span class="stat-trend-up">&uarr; Active Cohort</span>
            <span>across 4 Engineering Depts</span>
          </div>
        </div>

        <div class="stat-card" style="--accent-color: var(--emerald); --accent-light: var(--emerald-ligh
t);">
          <div class="stat-header">
            <span class="stat-title">Cohort Mean CGPA</span>
            <div class="stat-icon">
```

```
            <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
              <circle cx="12" cy="8" r="7"></circle>
              <polyline points="8.21 13.89 7 23 12 20 17 23 15.79 13.88"></polyline>
            </svg>
          </div>
        </div>
        <div class="stat-value" id="metric-avg-cgpa">8.82</div>
        <div class="stat-footer">
          <span class="stat-trend-up">&uarr; Comprehensive</span>
          <span>scale out of 10.0</span>
        </div>
      </div>

      <div class="stat-card" style="--accent-color: var(--cyan); --accent-light: var(--cyan-light);">
        <div class="stat-header">
          <span class="stat-title">Campus Placement Rate</span>
          <div class="stat-icon">
            <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
              <rect x="2" y="7" width="20" height="14" rx="2" ry="2"></rect>
              <path d="M16 21V5a2 2 0 0 0-2-2h-4a2 2 0 0 0-2 2v16"></path>
            </svg>
          </div>
        </div>
        <div class="stat-value" id="metric-placement-rate">60%</div>
        <div class="stat-footer">
          <span class="stat-trend-up">Tier-1 Offers</span>
          <span>Google, NVIDIA, Microsoft</span>
        </div>
      </div>

      <div class="stat-card" style="--accent-color: var(--purple); --accent-light: var(--purple-light);">
        <div class="stat-header">
          <span class="stat-title">Publications & Capstones</span>
          <div class="stat-icon">
            <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
              <path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"></path>
              <polyline points="14 2 14 8 20 8"></polyline>
              <line x1="16" y1="13" x2="8" y2="13"></line>
              <line x1="16" y1="17" x2="8" y2="17"></line>
              <polyline points="10 9 9 9 8 9"></polyline>
            </svg>
          </div>
        </div>
        <div class="stat-value" id="metric-publications">5</div>
        <div class="stat-footer">
          <span class="stat-trend-up">&uarr; Peer-reviewed</span>
          <span>IEEE & ACM venues</span>
        </div>
      </div>
    </div>

    <!-- Dynamic Content Body Rendered by app.js -->
    <div id="main-content"></div>

  </main>

  <!-- Modal Dialog Overlay Container -->
  <div class="modal-overlay" id="modal-overlay">
    <div class="modal-box">
      <div class="modal-header">
        <h3 id="modal-title">Modal Title</h3>
        <button class="modal-close-btn" id="modal-close-btn">&times;</button>
```

```
      </div>
      <div class="modal-body" id="modal-body">
        <!-- Injected dynamically -->
      </div>
      <div class="modal-footer" id="modal-footer">
        <!-- Injected dynamically -->
      </div>
    </div>
  </div>

  <!-- Toast Notification Container -->
  <div class="toast-container" id="toast-container"></div>

  <!-- Application Logic Scripts -->
  <script src="data.js"></script>
  <script src="app.js"></script>
</body>
</html>

```

#5 CSS styles.css

1,191 lines • 24.5 KB

Styling & Visual Design System (CSS3 Variables & Responsive Layout)

```css
/* ================================================================================
   Educational Institution Management System (IMS) - Core Design System
   Based on IEEE 10058674, IEEE 10516390, IEEE 7384005
   ================================================================================ */

:root {
  /* Color Palette - Curated Cyber Slate & Luminous Accents */
  --bg-primary: #0a0d14;
  --bg-secondary: #0f1523;
  --bg-card: rgba(18, 24, 38, 0.75);
  --bg-card-hover: rgba(28, 38, 60, 0.85);
  --bg-elevated: #161e31;
  --bg-glass: rgba(15, 23, 42, 0.65);

  --border-subtle: rgba(255, 255, 255, 0.08);
  --border-focus: rgba(99, 102, 241, 0.5);
  --border-glow: rgba(99, 102, 241, 0.25);

  --text-main: #f8fafc;
  --text-muted: #94a3b8;
  --text-dim: #64748b;
  --text-inverse: #0f172a;

  /* Accent Colors */
  --primary: #6366f1;
  --primary-hover: #4f46e5;
  --primary-light: rgba(99, 102, 241, 0.15);

  --cyan: #06b6d4;
  --cyan-light: rgba(6, 182, 212, 0.15);

  --emerald: #10b981;
  --emerald-light: rgba(16, 185, 129, 0.15);

  --amber: #f59e0b;
  --amber-light: rgba(245, 158, 11, 0.15);

  --rose: #f43f5e;
  --rose-light: rgba(244, 63, 94, 0.15);

  --purple: #a855f7;
  --purple-light: rgba(168, 85, 247, 0.15);

  /* Shadows & Radius */
  --radius-sm: 6px;
  --radius-md: 10px;
  --radius-lg: 16px;
  --radius-xl: 24px;

  --shadow-sm: 0 2px 4px rgba(0, 0, 0, 0.25);
  --shadow-md: 0 4px 16px -2px rgba(0, 0, 0, 0.4);
  --shadow-lg: 0 10px 30px -5px rgba(0, 0, 0, 0.6);
  --shadow-glow: 0 0 25px rgba(99, 102, 241, 0.25);

  --font-sans: 'Plus Jakarta Sans', system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans
-serif;
  --font-mono: 'JetBrains Mono', 'Fira Code', monospace;
  --transition-fast: 0.15s ease;
```

```
  --transition-normal: 0.25s cubic-bezier(0.4, 0, 0.2, 1);
}

/* Base Reset & Typography */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: var(--font-sans);
  background-color: var(--bg-primary);
  color: var(--text-main);
  line-height: 1.5;
  -webkit-font-smoothing: antialiased;
  min-height: 100vh;
  display: flex;
  flex-direction: column;
  overflow-x: hidden;
  background-image:
    radial-gradient(circle at 15% 15%, rgba(99, 102, 241, 0.08) 0%, transparent 40%),
    radial-gradient(circle at 85% 85%, rgba(6, 182, 212, 0.06) 0%, transparent 40%),
    linear-gradient(rgba(255, 255, 255, 0.015) 1px, transparent 1px),
    linear-gradient(90deg, rgba(255, 255, 255, 0.015) 1px, transparent 1px);
  background-size: 100% 100%, 100% 100%, 40px 40px, 40px 40px;
}

/* Custom Scrollbar */
::-webkit-scrollbar {
  width: 8px;
  height: 8px;
}
::-webkit-scrollbar-track {
  background: var(--bg-primary);
}
::-webkit-scrollbar-thumb {
  background: var(--border-subtle);
  border-radius: 4px;
}
::-webkit-scrollbar-thumb:hover {
  background: var(--text-dim);
}

/* App Header / Navigation */
.app-header {
  position: sticky;
  top: 0;
  z-index: 50;
  backdrop-filter: blur(16px);
  background: var(--bg-glass);
  border-bottom: 1px solid var(--border-subtle);
  padding: 0.75rem 1.5rem;
  display: flex;
  align-items: center;
  justify-content: space-between;
}

.brand-section {
  display: flex;
  align-items: center;
  gap: 0.85rem;
}

.brand-icon {
```

```
  width: 40px;
  height: 40px;
  border-radius: var(--radius-md);
  background: linear-gradient(135deg, var(--primary), var(--cyan));
  display: flex;
  align-items: center;
  justify-content: center;
  box-shadow: var(--shadow-glow);
  color: #fff;
  font-size: 1.25rem;
  font-weight: 700;
}

.brand-text h1 {
  font-size: 1.15rem;
  font-weight: 700;
  letter-spacing: -0.02em;
  background: linear-gradient(90deg, #ffffff, #cbd5e1);
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
}

.brand-text .subtitle {
  font-size: 0.75rem;
  color: var(--cyan);
  font-weight: 500;
  display: flex;
  align-items: center;
  gap: 0.4rem;
}

.brand-text .subtitle .badge-tag {
  background: rgba(6, 182, 212, 0.15);
  color: #38bdf8;
  padding: 0.1rem 0.4rem;
  border-radius: 4px;
  font-size: 0.65rem;
  font-family: var(--font-mono);
}

/* Role Selector Pill */
.header-actions {
  display: flex;
  align-items: center;
  gap: 1.25rem;
}

.role-switcher-container {
  display: flex;
  align-items: center;
  gap: 0.5rem;
  background: var(--bg-elevated);
  border: 1px solid var(--border-subtle);
  padding: 0.25rem 0.35rem;
  border-radius: 50px;
}

.role-label {
  font-size: 0.75rem;
  color: var(--text-muted);
  padding-left: 0.6rem;
  text-transform: uppercase;
  font-weight: 600;
  letter-spacing: 0.05em;
}
```

```
188
189 .role-btn {
190   background: transparent;
191   border: none;
192   color: var(--text-muted);
193   font-family: inherit;
194   font-size: 0.8rem;
195   font-weight: 600;
196   padding: 0.4rem 0.85rem;
197   border-radius: 50px;
198   cursor: pointer;
199   transition: all var(--transition-fast);
200   display: flex;
201   align-items: center;
202   gap: 0.4rem;
203 }
204
205 .role-btn:hover {
206   color: var(--text-main);
207 }
208
209 .role-btn.active {
210   background: linear-gradient(135deg, var(--primary), #4338ca);
211   color: #fff;
212   box-shadow: 0 2px 10px rgba(99, 102, 241, 0.4);
213 }
214
215 .btn-quick {
216   background: var(--bg-card);
217   border: 1px solid var(--border-subtle);
218   color: var(--text-main);
219   padding: 0.5rem 0.9rem;
220   border-radius: var(--radius-md);
221   font-size: 0.8rem;
222   font-weight: 600;
223   cursor: pointer;
224   display: inline-flex;
225   align-items: center;
226   gap: 0.4rem;
227   transition: all var(--transition-fast);
228 }
229
230 .btn-quick:hover {
231   background: var(--bg-elevated);
232   border-color: var(--primary);
233   transform: translateY(-1px);
234 }
235
236 .btn-primary {
237   background: linear-gradient(135deg, var(--primary), #4f46e5);
238   border: 1px solid rgba(255, 255, 255, 0.15);
239   color: #fff;
240   box-shadow: var(--shadow-sm);
241 }
242
243 .btn-primary:hover {
244   box-shadow: var(--shadow-glow);
245   background: linear-gradient(135deg, #4f46e5, #4338ca);
246 }
247
248 /* Layout Container */
249 .app-container {
250   max-width: 1540px;
251   margin: 0 auto;
252   padding: 1.5rem;
```

```
  width: 100%;
  flex: 1;
}

/* Secondary Navigation / Tabs */
.tabs-navigation {
  display: flex;
  align-items: center;
  gap: 0.5rem;
  border-bottom: 1px solid var(--border-subtle);
  margin-bottom: 1.5rem;
  overflow-x: auto;
  padding-bottom: 0.25rem;
}

.tab-nav-btn {
  background: transparent;
  border: none;
  color: var(--text-muted);
  font-family: inherit;
  font-size: 0.875rem;
  font-weight: 600;
  padding: 0.6rem 1rem;
  border-radius: var(--radius-md) var(--radius-md) 0 0;
  cursor: pointer;
  display: flex;
  align-items: center;
  gap: 0.5rem;
  position: relative;
  transition: color var(--transition-fast);
  white-space: nowrap;
}

.tab-nav-btn:hover {
  color: var(--text-main);
}

.tab-nav-btn.active {
  color: var(--primary);
}

.tab-nav-btn.active::after {
  content: '';
  position: absolute;
  bottom: -0.25rem;
  left: 0;
  right: 0;
  height: 2px;
  background: linear-gradient(90deg, var(--primary), var(--cyan));
  box-shadow: 0 0 10px var(--primary);
}

/* Banner / Overview Hero */
.hero-stats-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
  gap: 1rem;
  margin-bottom: 1.75rem;
}

.stat-card {
  background: var(--bg-card);
  border: 1px solid var(--border-subtle);
  border-radius: var(--radius-lg);
  padding: 1.25rem;
```

```
  backdrop-filter: blur(12px);
  position: relative;
  overflow: hidden;
  transition: transform var(--transition-fast), border-color var(--transition-fast);
}

.stat-card:hover {
  transform: translateY(-2px);
  border-color: rgba(99, 102, 241, 0.3);
}

.stat-card::before {
  content: '';
  position: absolute;
  top: 0;
  left: 0;
  width: 4px;
  height: 100%;
  background: var(--accent-color, var(--primary));
}

.stat-header {
  display: flex;
  align-items: center;
  justify-content: space-between;
  margin-bottom: 0.5rem;
}

.stat-title {
  font-size: 0.8rem;
  text-transform: uppercase;
  color: var(--text-muted);
  font-weight: 600;
  letter-spacing: 0.05em;
}

.stat-icon {
  width: 32px;
  height: 32px;
  border-radius: var(--radius-sm);
  background: var(--accent-light, var(--primary-light));
  color: var(--accent-color, var(--primary));
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 1rem;
}

.stat-value {
  font-size: 1.85rem;
  font-weight: 800;
  letter-spacing: -0.03em;
  color: var(--text-main);
  line-height: 1.1;
}

.stat-footer {
  margin-top: 0.4rem;
  font-size: 0.75rem;
  color: var(--text-muted);
  display: flex;
  align-items: center;
  gap: 0.35rem;
}

```

```
.stat-trend-up {
  color: var(--emerald);
  font-weight: 600;
}

/* Glass Card Container */
.glass-panel {
  background: var(--bg-card);
  border: 1px solid var(--border-subtle);
  border-radius: var(--radius-lg);
  backdrop-filter: blur(12px);
  padding: 1.5rem;
  box-shadow: var(--shadow-md);
  margin-bottom: 1.5rem;
}

.panel-header {
  display: flex;
  align-items: center;
  justify-content: space-between;
  margin-bottom: 1.25rem;
  flex-wrap: wrap;
  gap: 1rem;
}

.panel-title-group h2 {
  font-size: 1.25rem;
  font-weight: 700;
  color: var(--text-main);
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.panel-title-group p {
  font-size: 0.8rem;
  color: var(--text-muted);
  margin-top: 0.15rem;
}

/* Filter Controls Toolbar */
.toolbar-controls {
  display: flex;
  align-items: center;
  gap: 0.75rem;
  flex-wrap: wrap;
}

.search-input-wrapper {
  position: relative;
  min-width: 260px;
}

.search-input-wrapper input {
  width: 100%;
  background: var(--bg-secondary);
  border: 1px solid var(--border-subtle);
  border-radius: var(--radius-md);
  color: var(--text-main);
  font-family: inherit;
  font-size: 0.825rem;
  padding: 0.5rem 0.85rem 0.5rem 2.25rem;
  outline: none;
  transition: border-color var(--transition-fast);
}
```

```
448
449  .search-input-wrapper input:focus {
450    border-color: var(--primary);
451    box-shadow: 0 0 0 2px var(--border-glow);
452  }
453
454  .search-icon {
455    position: absolute;
456    left: 0.75rem;
457    top: 50%;
458    transform: translateY(-50%);
459    color: var(--text-dim);
460    pointer-events: none;
461  }
462
463  .select-filter {
464    background: var(--bg-secondary);
465    border: 1px solid var(--border-subtle);
466    border-radius: var(--radius-md);
467    color: var(--text-main);
468    font-family: inherit;
469    font-size: 0.825rem;
470    padding: 0.5rem 2rem 0.5rem 0.75rem;
471    outline: none;
472    cursor: pointer;
473    appearance: none;
474    background-image: url("data:image/svg+xml;charset=UTF-8,%3csvg xmlns='http://www.w3.org/2000/svg' viewBo
x='0 0 24 24' fill='none' stroke='%2394a3b8' stroke-width='2' stroke-linecap='round' stroke-linejoin='round'%3e%3cp
olyline points='6 9 12 15 18 9'%3e%3c/polyline%3e%3c/svg%3e");
475    background-repeat: no-repeat;
476    background-position: right 0.6rem center;
477    background-size: 14px;
478  }
479
480  .select-filter:focus {
481    border-color: var(--primary);
482  }
483
484  /* Modern Data Table */
485  .table-responsive {
486    width: 100%;
487    overflow-x: auto;
488    border-radius: var(--radius-md);
489    border: 1px solid var(--border-subtle);
490    background: var(--bg-secondary);
491  }
492
493  .data-table {
494    width: 100%;
495    border-collapse: collapse;
496    text-align: left;
497    font-size: 0.85rem;
498  }
499
500  .data-table th {
501    background: rgba(15, 23, 42, 0.95);
502    padding: 0.85rem 1rem;
503    color: var(--text-muted);
504    font-weight: 600;
505    text-transform: uppercase;
506    font-size: 0.7rem;
507    letter-spacing: 0.05em;
508    border-bottom: 1px solid var(--border-subtle);
509    white-space: nowrap;
510  }
```

```
511
512 .data-table td {
513   padding: 0.95rem 1rem;
514   border-bottom: 1px solid var(--border-subtle);
515   color: var(--text-main);
516   vertical-align: middle;
517 }
518
519 .data-table tr:last-child td {
520   border-bottom: none;
521 }
522
523 .data-table tbody tr {
524   transition: background-color var(--transition-fast);
525 }
526
527 .data-table tbody tr:hover {
528   background-color: rgba(99, 102, 241, 0.05);
529 }
530
531 /* Badges & Tags */
532 .badge {
533   display: inline-flex;
534   align-items: center;
535   gap: 0.35rem;
536   padding: 0.2rem 0.55rem;
537   border-radius: 50px;
538   font-size: 0.725rem;
539   font-weight: 600;
540   white-space: nowrap;
541 }
542
543 .badge-emerald {
544   background: var(--emerald-light);
545   color: #34d399;
546   border: 1px solid rgba(16, 185, 129, 0.3);
547 }
548
549 .badge-cyan {
550   background: var(--cyan-light);
551   color: #38bdf8;
552   border: 1px solid rgba(6, 182, 212, 0.3);
553 }
554
555 .badge-amber {
556   background: var(--amber-light);
557   color: #fbbf24;
558   border: 1px solid rgba(245, 158, 11, 0.3);
559 }
560
561 .badge-purple {
562   background: var(--purple-light);
563   color: #c084fc;
564   border: 1px solid rgba(168, 85, 247, 0.3);
565 }
566
567 .badge-rose {
568   background: var(--rose-light);
569   color: #fb7185;
570   border: 1px solid rgba(244, 63, 94, 0.3);
571 }
572
573 .student-identity {
574   display: flex;
575   align-items: center;
```

```
  gap: 0.75rem;
}

.student-avatar {
  width: 36px;
  height: 36px;
  border-radius: 50%;
  background: linear-gradient(135deg, #3730a3, #4338ca);
  color: #fff;
  display: flex;
  align-items: center;
  justify-content: center;
  font-weight: 700;
  font-size: 0.85rem;
  flex-shrink: 0;
  border: 2px solid var(--border-subtle);
}

.student-name {
  font-weight: 600;
  color: var(--text-main);
}

.student-roll {
  font-size: 0.75rem;
  color: var(--text-muted);
  font-family: var(--font-mono);
}

/* CGPA Progress Pill */
.cgpa-indicator {
  display: inline-flex;
  align-items: center;
  gap: 0.5rem;
}

.cgpa-pill {
  font-family: var(--font-mono);
  font-weight: 700;
  font-size: 0.85rem;
  padding: 0.15rem 0.5rem;
  border-radius: var(--radius-sm);
}

.cgpa-pill.high {
  background: rgba(16, 185, 129, 0.2);
  color: #34d399;
}
.cgpa-pill.med {
  background: rgba(6, 182, 212, 0.2);
  color: #38bdf8;
}
.cgpa-pill.low {
  background: rgba(245, 158, 11, 0.2);
  color: #fbbf24;
}

/* Table Actions */
.action-btn-group {
  display: flex;
  align-items: center;
  gap: 0.35rem;
}

.action-icon-btn {
```

```
  background: transparent;
  border: 1px solid transparent;
  color: var(--text-muted);
  width: 30px;
  height: 30px;
  border-radius: var(--radius-sm);
  display: flex;
  align-items: center;
  justify-content: center;
  cursor: pointer;
  transition: all var(--transition-fast);
}

.action-icon-btn:hover {
  background: var(--bg-elevated);
  border-color: var(--border-subtle);
  color: var(--text-main);
}

.action-icon-btn.delete:hover {
  background: var(--rose-light);
  color: var(--rose);
  border-color: rgba(244, 63, 94, 0.3);
}

/* 4-Tier Architecture Viewer (IEEE 7384005) */
.architecture-grid {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 1rem;
  margin-top: 1rem;
}

.tier-column {
  background: var(--bg-secondary);
  border: 1px solid var(--border-subtle);
  border-radius: var(--radius-md);
  padding: 1.25rem;
  position: relative;
  display: flex;
  flex-direction: column;
}

.tier-column.active-tier {
  border-color: var(--primary);
  box-shadow: 0 0 15px rgba(99, 102, 241, 0.15);
}

.tier-header {
  border-bottom: 1px solid var(--border-subtle);
  padding-bottom: 0.75rem;
  margin-bottom: 1rem;
}

.tier-tag {
  font-size: 0.65rem;
  font-family: var(--font-mono);
  color: var(--cyan);
  text-transform: uppercase;
  font-weight: 700;
  letter-spacing: 0.05em;
}

.tier-title {
  font-size: 1rem;
```

```
  font-weight: 700;
  color: var(--text-main);
  margin-top: 0.2rem;
}

.tier-component-list {
  display: flex;
  flex-direction: column;
  gap: 0.6rem;
  flex: 1;
}

.tier-box {
  background: var(--bg-elevated);
  border: 1px solid var(--border-subtle);
  padding: 0.75rem;
  border-radius: var(--radius-sm);
  font-size: 0.8rem;
  transition: transform var(--transition-fast);
}

.tier-box:hover {
  border-color: var(--primary-hover);
  transform: translateX(3px);
}

.tier-box .box-title {
  font-weight: 600;
  color: #fff;
  display: flex;
  align-items: center;
  justify-content: space-between;
}

.tier-box .box-desc {
  font-size: 0.725rem;
  color: var(--text-muted);
  margin-top: 0.3rem;
}

/* ER Diagram Interactive Canvas */
.er-container {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
  gap: 1.25rem;
  margin-top: 1.25rem;
}

.er-card {
  background: var(--bg-secondary);
  border: 1px solid var(--border-subtle);
  border-radius: var(--radius-md);
  overflow: hidden;
  transition: border-color var(--transition-fast);
}

.er-card:hover {
  border-color: var(--cyan);
}

.er-card-header {
  background: var(--bg-elevated);
  padding: 0.75rem 1rem;
  font-weight: 700;
  font-size: 0.85rem;
```

```
    color: #38bdf8;
    display: flex;
    align-items: center;
    justify-content: space-between;
    border-bottom: 1px solid var(--border-subtle);
  }

  .er-card-body {
    padding: 0.5rem 0;
  }

  .er-field {
    display: flex;
    align-items: center;
    justify-content: space-between;
    padding: 0.35rem 1rem;
    font-size: 0.75rem;
    border-bottom: 1px solid rgba(255, 255, 255, 0.03);
  }

  .er-field:last-child {
    border-bottom: none;
  }

  .er-field-name {
    color: var(--text-main);
    font-family: var(--font-mono);
  }

  .er-field-meta {
    display: flex;
    align-items: center;
    gap: 0.4rem;
  }

  .pk-tag {
    background: rgba(245, 158, 11, 0.2);
    color: #f59e0b;
    font-size: 0.6rem;
    font-weight: 700;
    padding: 0.1rem 0.3rem;
    border-radius: 3px;
  }

  .fk-tag {
    background: rgba(99, 102, 241, 0.2);
    color: #818cf8;
    font-size: 0.6rem;
    font-weight: 700;
    padding: 0.1rem 0.3rem;
    border-radius: 3px;
  }

  .er-field-type {
    color: var(--text-dim);
    font-size: 0.7rem;
  }

  /* Modal Dialog */
  .modal-overlay {
    position: fixed;
    inset: 0;
    background: rgba(5, 8, 15, 0.85);
    backdrop-filter: blur(8px);
    z-index: 100;
```

```
  display: flex;
  align-items: center;
  justify-content: center;
  padding: 1.5rem;
  opacity: 0;
  pointer-events: none;
  transition: opacity var(--transition-normal);
}

.modal-overlay.active {
  opacity: 1;
  pointer-events: auto;
}

.modal-box {
  background: var(--bg-card);
  border: 1px solid var(--border-subtle);
  border-radius: var(--radius-xl);
  width: 100%;
  max-width: 760px;
  max-height: 90vh;
  display: flex;
  flex-direction: column;
  overflow: hidden;
  box-shadow: var(--shadow-lg), var(--shadow-glow);
  transform: translateY(20px) scale(0.98);
  transition: transform var(--transition-normal);
}

.modal-overlay.active .modal-box {
  transform: translateY(0) scale(1);
}

.modal-header {
  padding: 1.25rem 1.5rem;
  border-bottom: 1px solid var(--border-subtle);
  display: flex;
  align-items: center;
  justify-content: space-between;
}

.modal-header h3 {
  font-size: 1.15rem;
  font-weight: 700;
  color: var(--text-main);
}

.modal-close-btn {
  background: transparent;
  border: none;
  color: var(--text-muted);
  font-size: 1.35rem;
  cursor: pointer;
  line-height: 1;
}

.modal-close-btn:hover {
  color: #fff;
}

.modal-body {
  padding: 1.5rem;
  overflow-y: auto;
  flex: 1;
}
```

```
901
902 .modal-footer {
903   padding: 1rem 1.5rem;
904   border-top: 1px solid var(--border-subtle);
905   display: flex;
906   align-items: center;
907   justify-content: flex-end;
908   gap: 0.75rem;
909   background: var(--bg-elevated);
910 }
911
912 /* Form Controls */
913 .form-grid {
914   display: grid;
915   grid-template-columns: repeat(2, 1fr);
916   gap: 1.25rem;
917 }
918
919 .form-group {
920   display: flex;
921   flex-direction: column;
922   gap: 0.4rem;
923 }
924
925 .form-group.col-span-2 {
926   grid-column: span 2;
927 }
928
929 .form-label {
930   font-size: 0.775rem;
931   font-weight: 600;
932   color: var(--text-muted);
933 }
934
935 .form-control {
936   background: var(--bg-secondary);
937   border: 1px solid var(--border-subtle);
938   border-radius: var(--radius-md);
939   color: var(--text-main);
940   font-family: inherit;
941   font-size: 0.85rem;
942   padding: 0.6rem 0.85rem;
943   outline: none;
944   transition: border-color var(--transition-fast);
945 }
946
947 .form-control:focus {
948   border-color: var(--primary);
949   box-shadow: 0 0 0 2px var(--border-glow);
950 }
951
952 textarea.form-control {
953   min-height: 80px;
954   resize: vertical;
955 }
956
957 /* Detail Dossier View */
958 .dossier-grid {
959   display: grid;
960   grid-template-columns: 280px 1fr;
961   gap: 1.5rem;
962 }
963
964 .dossier-sidebar {
965   background: var(--bg-secondary);
```

```
  border: 1px solid var(--border-subtle);
  border-radius: var(--radius-lg);
  padding: 1.5rem;
  display: flex;
  flex-direction: column;
  align-items: center;
  text-align: center;
}

.dossier-avatar-large {
  width: 90px;
  height: 90px;
  border-radius: 50%;
  background: linear-gradient(135deg, var(--primary), var(--purple));
  color: #fff;
  font-size: 2.25rem;
  font-weight: 800;
  display: flex;
  align-items: center;
  justify-content: center;
  margin-bottom: 1rem;
  box-shadow: 0 4px 15px rgba(99, 102, 241, 0.35);
  border: 3px solid var(--border-subtle);
}

.dossier-info-list {
  width: 100%;
  margin-top: 1.25rem;
  text-align: left;
}

.dossier-info-item {
  padding: 0.5rem 0;
  border-bottom: 1px solid var(--border-subtle);
  font-size: 0.775rem;
  display: flex;
  justify-content: space-between;
}

.dossier-info-item:last-child {
  border-bottom: none;
}

.dossier-info-item .k {
  color: var(--text-muted);
}
.dossier-info-item .v {
  color: var(--text-main);
  font-weight: 600;
}

.dossier-section-title {
  font-size: 0.95rem;
  font-weight: 700;
  color: var(--text-main);
  margin-bottom: 0.75rem;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.item-chip-list {
  display: flex;
  flex-direction: column;
  gap: 0.6rem;
```

```
  margin-bottom: 1.25rem;
}

.item-chip {
  background: var(--bg-secondary);
  border: 1px solid var(--border-subtle);
  border-radius: var(--radius-md);
  padding: 0.85rem;
}

.item-chip .chip-head {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 0.25rem;
}

.item-chip .chip-title {
  font-size: 0.85rem;
  font-weight: 600;
  color: #fff;
}

.item-chip .chip-sub {
  font-size: 0.75rem;
  color: var(--text-muted);
}

/* Chart Canvas Section */
.charts-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(360px, 1fr));
  gap: 1.25rem;
  margin-bottom: 1.5rem;
}

.chart-card {
  background: var(--bg-card);
  border: 1px solid var(--border-subtle);
  border-radius: var(--radius-lg);
  padding: 1.25rem;
}

.chart-header {
  font-size: 0.9rem;
  font-weight: 700;
  color: var(--text-main);
  margin-bottom: 1rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.bar-chart-container {
  display: flex;
  flex-direction: column;
  gap: 0.75rem;
}

.bar-row {
  display: flex;
  flex-direction: column;
  gap: 0.25rem;
}

```

```
.bar-row-header {
  display: flex;
  justify-content: space-between;
  font-size: 0.75rem;
  color: var(--text-muted);
}

.bar-track {
  height: 8px;
  background: var(--bg-elevated);
  border-radius: 4px;
  overflow: hidden;
}

.bar-fill {
  height: 100%;
  border-radius: 4px;
  background: linear-gradient(90deg, var(--primary), var(--cyan));
  transition: width 0.6s ease;
}

/* Toast Notifications */
.toast-container {
  position: fixed;
  bottom: 1.5rem;
  right: 1.5rem;
  z-index: 200;
  display: flex;
  flex-direction: column;
  gap: 0.5rem;
}

.toast {
  background: var(--bg-elevated);
  border: 1px solid var(--border-subtle);
  border-radius: var(--radius-md);
  padding: 0.85rem 1.25rem;
  font-size: 0.825rem;
  color: #fff;
  display: flex;
  align-items: center;
  gap: 0.6rem;
  box-shadow: var(--shadow-lg);
  animation: slideIn 0.3s cubic-bezier(0.16, 1, 0.3, 1) forwards;
}

.toast.success {
  border-color: rgba(16, 185, 129, 0.4);
}
.toast.info {
  border-color: rgba(6, 182, 212, 0.4);
}

@keyframes slideIn {
  from {
    transform: translateX(100%);
    opacity: 0;
  }
  to {
    transform: translateX(0);
    opacity: 1;
  }
}

/* Responsive adjustments */
```

```
@media (max-width: 1024px) {
  .architecture-grid {
    grid-template-columns: repeat(2, 1fr);
  }
  .dossier-grid {
    grid-template-columns: 1fr;
  }
}

@media (max-width: 768px) {
  .app-header {
    flex-direction: column;
    gap: 1rem;
    align-items: flex-start;
  }
  .header-actions {
    width: 100%;
    justify-content: space-between;
    flex-wrap: wrap;
  }
  .architecture-grid {
    grid-template-columns: 1fr;
  }
  .form-grid {
    grid-template-columns: 1fr;
  }
  .form-group.col-span-2 {
    grid-column: span 1;
  }
}

```

#6 JAVASCRIPT **data.js** 244 lines • 11.2 KB

Data Layer (Tier 4) - Seed Data & Entity Schemas

```
/**
 * Educational Institution Management Information System (IMS) - Normalized Relational Data
 * Based on Schema Specifications from:
 * - IEEE 7384005: ER Entities (Students, Teachers, Depts, Majors, Classes, Courses, Scores, Placements)
 * - IEEE 10058674: Student Module (Admissions, Promotions, Research Papers, Internships, Events)
 * - IEEE 10516390: Multi-role communication & central database
 */

const IMS_INITIAL_DATA = {
  departments: [
    { id: 'dept-cse', code: 'CSE', name: 'Computer Science and Engineering', head: 'Dr. Aris Thorne' },
    { id: 'dept-aids', code: 'AIDS', name: 'Artificial Intelligence & Data Science', head: 'Dr. Elena Rostova' },
    { id: 'dept-ece', code: 'ECE', name: 'Electronics and Communication', head: 'Dr. Marcus Vance' },
    { id: 'dept-me', code: 'ME', name: 'Mechanical & Mechatronics Engineering', head: 'Dr. Sarah Lin' }
  ],

  majors: [
    { id: 'maj-cse-core', deptId: 'dept-cse', name: 'B.Tech Computer Science', degree: 'Undergraduate' },
    { id: 'maj-ai-ds', deptId: 'dept-aids', name: 'B.Tech AI & Data Analytics', degree: 'Undergraduate' },
    { id: 'maj-ece-iot', deptId: 'dept-ece', name: 'B.Tech Embedded Systems & IoT', degree: 'Undergraduate' },
    { id: 'maj-robotics', deptId: 'dept-me', name: 'B.Tech Robotics & Automation', degree: 'Undergraduate' }
  ],

  classes: [
    { id: 'cls-cse-2023', name: 'CSE-Alpha (Batch 2023-2027)', currentSemester: 6, counselorId: 't-101' },
    { id: 'cls-aids-2023', name: 'AIDS-Prime (Batch 2023-2027)', currentSemester: 6, counselorId: 't-102' },
    { id: 'cls-ece-2024', name: 'ECE-Omega (Batch 2024-2028)', currentSemester: 4, counselorId: 't-103' },
    { id: 'cls-cse-2022', name: 'CSE-Senior (Batch 2022-2026)', currentSemester: 8, counselorId: 't-101' }
  ],

  teachers: [
    { id: 't-101', name: 'Prof. Donald Knuth Jr.', department: 'CSE', email: 'd.knuth@ims.edu', title: 'Senior Professor' },
    { id: 't-102', name: 'Dr. Barbara Liskov', department: 'AIDS', email: 'b.liskov@ims.edu', title: 'Associate Professor' },
    { id: 't-103', name: 'Dr. Claude Shannon', department: 'ECE', email: 'c.shannon@ims.edu', title: 'Department Chair' },
    { id: 't-104', name: 'Dr. Ada Lovelace', department: 'CSE', email: 'a.lovelace@ims.edu', title: 'Assistant Professor' }
  ],

  courses: [
    { id: 'c-301', code: 'CS301', name: 'Distributed Systems & Cloud Computing', credits: 4, type: 'Core', dept: 'CSE', instructorId: 't-101' },
    { id: 'c-302', code: 'AI302', name: 'Deep Learning & Neural Architectures', credits: 4, type: 'Core', dept: 'AIDS', instructorId: 't-102' },
    { id: 'c-303', code: 'CS303', name: 'Database Internal Engine Design', credits: 3, type: 'Core', dept: 'CSE', instructorId: 't-104' },
    { id: 'c-304', code: 'EC204', name: 'Microcontrollers & Cyber-Physical Systems', credits: 4, type: 'Lab & Theory', dept: 'ECE', instructorId: 't-103' },
    { id: 'c-305', code: 'OE305', name: 'Technology Entrepreneurship & Ethics', credits: 2, type: 'Open Elective', dept: 'CSE', instructorId: 't-101' }
  ],

```

```
    students: [
      {
        id: 'stu-1001',
        rollNo: '2023CSE001',
        name: 'Devin Vance',
        email: 'devin.vance@student.ims.edu',
        deptId: 'dept-cse',
        majorId: 'maj-cse-core',
        classId: 'cls-cse-2023',
        admissionYear: 2023,
        currentSemester: 6,
        cgpa: 9.42,
        totalCreditsEarned: 132,
        academicStatus: 'Regular',
        promotionStatus: 'Promoted to Sem 6',
        transferHistory: 'Direct Admission via Merit Rank #142',
        scores: [
          { courseCode: 'CS301', courseName: 'Distributed Systems', internal: 28, assignments: 19, exam: 48, total: 95, grade: 'O', gpa: 10.0 },
          { courseCode: 'AI302', courseName: 'Deep Learning', internal: 27, assignments: 18, exam: 45, total: 90, grade: 'A+', gpa: 9.0 },
          { courseCode: 'CS303', courseName: 'Database Engines', internal: 29, assignments: 20, exam: 47, total: 96, grade: 'O', gpa: 10.0 }
        ],
        placement: {
          status: 'Placed',
          company: 'NVIDIA AI Labs',
          role: 'Systems Software Engineer Intern / FTE',
          ctc: '32.5 LPA',
          offerDate: '2026-03-15'
        },
        events: [
          { name: 'IEEE International Student Hackathon 2025', type: 'Hackathon', outcome: 'First Place Winner ($5,000)' },
          { name: 'Kaggle Grandmaster Summit', type: 'Workshop', outcome: 'Certificate of Excellence' }
        ],
        projectsAndPublications: [
          { title: 'High-Throughput Vector Indexing for Distributed RAG', type: 'Research Paper', venue: 'IEEE Trans. on Knowledge & Data Eng. (Under Review)', guide: 'Prof. Donald Knuth Jr.' },
          { title: 'Autonomous Drone Swarm Telemetry System', type: 'Capstone Project', venue: 'Annual Tech Expo 2025', guide: 'Dr. Claude Shannon' }
        ]
      },
      {
        id: 'stu-1002',
        rollNo: '2023AID014',
        name: 'Aria Chen',
        email: 'aria.chen@student.ims.edu',
        deptId: 'dept-aids',
        majorId: 'maj-ai-ds',
        classId: 'cls-aids-2023',
        admissionYear: 2023,
        currentSemester: 6,
        cgpa: 9.15,
        totalCreditsEarned: 128,
        academicStatus: 'Regular',
        promotionStatus: 'Promoted to Sem 6',
        transferHistory: 'Direct Admission via All India Exam Rank #310',
        scores: [
          { courseCode: 'AI302', courseName: 'Deep Learning', internal: 30, assignments: 20, exam: 48, total: 98, grade: 'O', gpa: 10.0 },
          { courseCode: 'CS301', courseName: 'Distributed Systems', internal: 26, assignments: 17, exam: 44, total: 87, grade: 'A', gpa: 8.5 }
        ],
        placement: {
```

```
          status: 'Placed',
          company: 'Google DeepMind Research',
          role: 'Research Associate Intern',
          ctc: '38.0 LPA',
          offerDate: '2026-02-10'
        },
        events: [
          { name: 'NeurIPS Workshop on Foundation Models', type: 'Conference', outcome: 'Poster Presentatio
n' },
          { name: 'ACM ICPC Regional Semifinals', type: 'Coding Contest', outcome: 'Rank 12' }
        ],
        projectsAndPublications: [
          { title: 'Self-Supervised Contrastive Representation for Medical Imaging', type: 'Published Pape
r', venue: 'IEEE CVPR Workshop 2025', guide: 'Dr. Barbara Liskov' }
        ]
      },
      {
        id: 'stu-1003',
        rollNo: '2022CSE089',
        name: 'Kavita Patel',
        email: 'kavita.p@student.ims.edu',
        deptId: 'dept-cse',
        majorId: 'maj-cse-core',
        classId: 'cls-cse-2022',
        admissionYear: 2022,
        currentSemester: 8,
        cgpa: 8.78,
        totalCreditsEarned: 168,
        academicStatus: 'Graduating Senior',
        promotionStatus: 'Promoted to Final Sem 8',
        transferHistory: 'Internal Branch Transfer from ECE in Sem 3 (Merit CGPA 9.2)',
        scores: [
          { courseCode: 'CS301', courseName: 'Distributed Systems', internal: 25, assignments: 18, exam: 42,
total: 85, grade: 'A', gpa: 8.5 },
          { courseCode: 'OE305', courseName: 'Tech Entrepreneurship', internal: 29, assignments: 19, exam: 4
6, total: 94, grade: 'O', gpa: 10.0 }
        ],
        placement: {
          status: 'Placed',
          company: 'Microsoft Azure Core',
          role: 'Software Development Engineer 1',
          ctc: '28.0 LPA',
          offerDate: '2025-11-20'
        },
        events: [
          { name: 'Microsoft Imagine Cup National Finals', type: 'Competition', outcome: '2nd Runner Up' }
        ],
        projectsAndPublications: [
          { title: 'Decentralized Identity Verification for University Credentials', type: 'Capstone Projec
t', venue: 'IMS Campus Deployment', guide: 'Dr. Ada Lovelace' }
        ]
      },
      {
        id: 'stu-1004',
        rollNo: '2024ECE045',
        name: 'Rohan Deshmukh',
        email: 'rohan.d@student.ims.edu',
        deptId: 'dept-ece',
        majorId: 'maj-ece-iot',
        classId: 'cls-ece-2024',
        admissionYear: 2024,
        currentSemester: 4,
        cgpa: 7.85,
        totalCreditsEarned: 84,
        academicStatus: 'Regular',
```

```
        promotionStatus: 'Promoted to Sem 4',
        transferHistory: 'Lateral Entry Admission',
        scores: [
          { courseCode: 'EC204', courseName: 'Microcontrollers & CPS', internal: 24, assignments: 16, exam: 39, total: 79, grade: 'B+', gpa: 7.5 }
        ],
        placement: {
          status: 'Eligible',
          company: 'Texas Instruments',
          role: 'Hardware Firmware Engineering Trainee',
          ctc: 'Pending Interview Round 2',
          offerDate: '—'
        },
        events: [
          { name: 'National Embedded Systems Expo', type: 'Exhibition', outcome: 'Best Hardware Prototype' }
        ],
        projectsAndPublications: [
          { title: 'Sub-gigahertz Mesh Network for Campus Emergency Beaconing', type: 'Mini Project', venue: 'Department Tech Day', guide: 'Dr. Claude Shannon' }
        ]
      },
      {
        id: 'stu-1005',
        rollNo: '2023CSE052',
        name: 'Sophia Martinez',
        email: 's.martinez@student.ims.edu',
        deptId: 'dept-cse',
        majorId: 'maj-cse-core',
        classId: 'cls-cse-2023',
        admissionYear: 2023,
        currentSemester: 6,
        cgpa: 8.92,
        totalCreditsEarned: 126,
        academicStatus: 'Regular',
        promotionStatus: 'Promoted to Sem 6',
        transferHistory: 'State Counseling Direct Admission',
        scores: [
          { courseCode: 'CS301', courseName: 'Distributed Systems', internal: 27, assignments: 18, exam: 45, total: 90, grade: 'A+', gpa: 9.0 },
          { courseCode: 'CS303', courseName: 'Database Engines', internal: 26, assignments: 17, exam: 43, total: 86, grade: 'A', gpa: 8.5 }
        ],
        placement: {
          status: 'Interviewing',
          company: 'Amazon AWS',
          role: 'Cloud Support / SDE Intern',
          ctc: 'Expected 24 LPA',
          offerDate: '—'
        },
        events: [
          { name: 'AWS Cloud Practitioner Bootcamp', type: 'Certification', outcome: 'Certified AWS Solutions Architect' }
        ],
        projectsAndPublications: [
          { title: 'Multi-Tenant Serverless Orchestration for Higher Education SIS', type: 'Research Paper', venue: 'IEEE CloudNet 2025', guide: 'Prof. Donald Knuth Jr.' }
        ]
      }
    ],

    announcements: [
      {
        id: 'ann-1',
        title: 'Spring 2026 Comprehensive Semester Grade Publication',
        author: 'Office of Controller of Examinations',
```

```
        date: '2026-04-10',
        priority: 'High',
        content: 'Grades for all B.Tech and M.Tech core and elective modules are now finalized in the Central Database layer. Students can view transcripts and review score breakdowns.'
      },
      {
        id: 'ann-2',
        title: 'Campus Recruitment Drive - Phase 2 Registration Open',
        author: 'Placement & Training Directorate',
        date: '2026-04-15',
        priority: 'Important',
        content: 'Eligible students with CGPA >= 7.5 and no active backlogs are invited to register for upcoming placement drives by Microsoft, NVIDIA, and Cisco.'
      },
      {
        id: 'ann-3',
        title: 'Call for Papers: IEEE Student Technical Symposium',
        author: 'Dean of Research & Sponsored Projects',
        date: '2026-04-18',
        priority: 'Normal',
        content: 'Submissions are invited for student capstone and research papers. Selected works will receive institutional publication funding.'
      }
    ]
  };

```

#7 JAVASCRIPT **app.js** 1,729 lines • 76.6 KB

Business Logic Layer (Tier 2) & Data Access Layer (Tier 3)

```
/**
 * Educational Institution Management Information System (IMS) - Application Controller
 * Realizing the 4-Tier Architecture from IEEE 7384005:
 *   [1] Web Display Layer
 *   [2] Business Logic Layer
 *   [3] Data Access Layer
 *   [4] Database Layer
 * Incorporating features from IEEE 10058674 (Student Module) & IEEE 10516390 (IMS Web Platform)
 */

// ============================================================================
// [LAYER 4 & 3] DATA ACCESS LAYER (DAL) & STORAGE
// ============================================================================
class DataAccessLayer {
  constructor() {
    this.STORAGE_KEY = 'IMS_EDUCATIONAL_SYSTEM_DATA_V1';
    this.state = this.loadState();
  }

  loadState() {
    try {
      if (typeof localStorage !== 'undefined' && localStorage) {
        const saved = localStorage.getItem(this.STORAGE_KEY);
        if (saved) {
          return JSON.parse(saved);
        }
      }
    } catch (e) {
      console.warn('Storage read fallback:', e);
    }
    // Deep clone initial data
    if (typeof IMS_INITIAL_DATA !== 'undefined') {
      return JSON.parse(JSON.stringify(IMS_INITIAL_DATA));
    }
    return { departments: [], majors: [], classes: [], teachers: [], courses: [], students: [], announcements: [] };
  }

  saveState() {
    try {
      if (typeof localStorage !== 'undefined' && localStorage) {
        localStorage.setItem(this.STORAGE_KEY, JSON.stringify(this.state));
      }
    } catch (e) {
      console.error('Storage write error:', e);
    }
  }

  resetToDefaults() {
    this.state = JSON.parse(JSON.stringify(IMS_INITIAL_DATA));
    this.saveState();
  }

  // Entities CRUD
  getStudents() {
    return this.state.students;
  }

```

```
  getStudentById(id) {
    return this.state.students.find(s => s.id === id);
  }

  createStudent(studentData) {
    studentData.id = 'stu-' + Date.now().toString().slice(-6);
    this.state.students.unshift(studentData);
    this.saveState();
    return studentData;
  }

  updateStudent(id, studentData) {
    const idx = this.state.students.findIndex(s => s.id === id);
    if (idx !== -1) {
      this.state.students[idx] = { ...this.state.students[idx], ...studentData, id };
      this.saveState();
      return this.state.students[idx];
    }
    return null;
  }

  deleteStudent(id) {
    const idx = this.state.students.findIndex(s => s.id === id);
    if (idx !== -1) {
      const removed = this.state.students.splice(idx, 1)[0];
      this.saveState();
      return removed;
    }
    return null;
  }

  getDepartments() { return this.state.departments; }
  getMajors() { return this.state.majors; }
  getClasses() { return this.state.classes; }
  getCourses() { return this.state.courses; }
  getTeachers() { return this.state.teachers; }
  getAnnouncements() { return this.state.announcements; }

  addAnnouncement(announcement) {
    announcement.id = 'ann-' + Date.now().toString().slice(-4);
    this.state.announcements.unshift(announcement);
    this.saveState();
    return announcement;
  }
}

// ============================================================================
// [LAYER 2] BUSINESS LOGIC LAYER (BLL)
// ============================================================================
class BusinessLogicLayer {
  constructor(dal) {
    this.dal = dal;
  }

  // Calculate Grade Letter and GPA from total score (IEEE 7384005 Comprehensive Score)
  calculateGradeFromTotal(total) {
    if (total >= 90) return { grade: 'O', gpa: 10.0 };
    if (total >= 80) return { grade: 'A+', gpa: 9.0 };
    if (total >= 70) return { grade: 'A', gpa: 8.0 };
    if (total >= 60) return { grade: 'B+', gpa: 7.0 };
    if (total >= 50) return { grade: 'B', gpa: 6.0 };
    if (total >= 40) return { grade: 'C', gpa: 5.0 };
    return { grade: 'F', gpa: 0.0 };
  }

```

```
    // Compute student cumulative CGPA dynamically from course scores
    computeStudentCGPA(student) {
      if (!student.scores || student.scores.length === 0) return student.cgpa || 0.0;
      const totalGpa = student.scores.reduce((acc, curr) => acc + (curr.gpa || 0), 0);
      const computed = (totalGpa / student.scores.length).toFixed(2);
      return parseFloat(computed);
    }

    // Verify campus placement eligibility (IEEE 10058674 Placement criteria)
    checkPlacementEligibility(student) {
      const cgpa = student.cgpa || 0;
      const hasBacklog = (student.scores || []).some(s => s.grade === 'F');
      if (cgpa >= 7.5 && !hasBacklog) {
        return { eligible: true, statusText: 'Eligible for Tier-1 Companies' };
      } else if (cgpa >= 6.0 && !hasBacklog) {
        return { eligible: true, statusText: 'Eligible for General Drives' };
      }
      return { eligible: false, statusText: 'Remediation Required (CGPA < 6.0 or Backlog)' };
    }

    // Process academic progression / student promotion (IEEE 10058674 Student Module)
    promoteStudentSemester(studentId) {
      const student = this.dal.getStudentById(studentId);
      if (!student) return { success: false, message: 'Student not found.' };

      if (student.currentSemester >= 8) {
        student.academicStatus = 'Graduated / Degree Awarded';
        student.promotionStatus = 'Degree Conferred (Class of ' + (student.admissionYear + 4) + ')';
      } else {
        student.currentSemester += 1;
        student.promotionStatus = `Successfully Promoted to Semester ${student.currentSemester}`;
        student.totalCreditsEarned = (student.totalCreditsEarned || 0) + 24;
      }
      this.dal.saveState();
      return { success: true, student, message: `Student promoted to Semester ${student.currentSemester}!`
};
    }

    // Compute institutional summary statistics
    getInstitutionalMetrics() {
      const students = this.dal.getStudents();
      const total = students.length;
      if (total === 0) {
        return { total: 0, avgCgpa: '0.00', placedCount: 0, placementRate: '0%', publicationsCount: 0 };
      }

      const totalCgpa = students.reduce((acc, s) => acc + (s.cgpa || 0), 0);
      const avgCgpa = (totalCgpa / total).toFixed(2);
      const placedCount = students.filter(s => s.placement && s.placement.status === 'Placed').length;
      const placementRate = Math.round((placedCount / total) * 100) + '%';

      let publicationsCount = 0;
      students.forEach(s => {
        if (s.projectsAndPublications) {
          publicationsCount += s.projectsAndPublications.length;
        }
      });

      return {
        total,
        avgCgpa,
        placedCount,
        placementRate,
        publicationsCount
      };
```

```
    }

    // Filter students by search term, department, and academic status
    filterStudents(searchTerm = '', deptId = 'all', status = 'all') {
      let list = this.dal.getStudents();

      if (deptId !== 'all') {
        list = list.filter(s => s.deptId === deptId);
      }

      if (status !== 'all') {
        if (status === 'Placed') {
          list = list.filter(s => s.placement && s.placement.status === 'Placed');
        } else if (status === 'Eligible') {
          list = list.filter(s => s.placement && (s.placement.status === 'Eligible' || s.placement.status ==
= 'Placed'));
        } else {
          list = list.filter(s => s.academicStatus === status);
        }
      }

      if (searchTerm.trim()) {
        const q = searchTerm.toLowerCase();
        list = list.filter(s =>
          (s.name && s.name.toLowerCase().includes(q)) ||
          (s.rollNo && s.rollNo.toLowerCase().includes(q)) ||
          (s.email && s.email.toLowerCase().includes(q))
        );
      }

      return list;
    }
  }

  // ============================================================================
  // [LAYER 1] WEB DISPLAY LAYER & UI CONTROLLER
  // ============================================================================
  class InstitutionManagementApp {
    constructor() {
      this.dal = new DataAccessLayer();
      this.bll = new BusinessLogicLayer(this.dal);

      // Current session state
      this.currentRole = 'admin'; // 'admin' | 'faculty' | 'student'
      this.activeTab = 'directory'; // 'directory' | 'academics' | 'placements' | 'research' | 'architectur
e' | 'communications'
      this.selectedStudentForDetail = null;

      this.initDOMReferences();
      this.bindEvents();
      this.render();
    }

    initDOMReferences() {
      // Header & Role switchers
      this.roleButtons = document.querySelectorAll('.role-btn');
      this.tabButtons = document.querySelectorAll('.tab-nav-btn');
      this.activeRoleBadge = document.getElementById('active-role-badge');

      // Metrics elements
      this.metricTotalStudents = document.getElementById('metric-total-students');
      this.metricAvgCgpa = document.getElementById('metric-avg-cgpa');
      this.metricPlacementRate = document.getElementById('metric-placement-rate');
      this.metricPublications = document.getElementById('metric-publications');

```

```
    // Content container & views
    this.mainContent = document.getElementById('main-content');

    // Modals
    this.modalOverlay = document.getElementById('modal-overlay');
    this.modalTitle = document.getElementById('modal-title');
    this.modalBody = document.getElementById('modal-body');
    this.modalFooter = document.getElementById('modal-footer');
    this.modalCloseBtn = document.getElementById('modal-close-btn');

    // Quick Action Buttons
    this.btnNewStudent = document.getElementById('btn-new-student');
    this.btnExportData = document.getElementById('btn-export-data');
    this.btnResetData = document.getElementById('btn-reset-data');
  }

  bindEvents() {
    // Role switcher clicks
    this.roleButtons.forEach(btn => {
      btn.addEventListener('click', (e) => {
        const role = e.currentTarget.dataset.role;
        this.setRole(role);
      });
    });

    // Tab navigation clicks
    this.tabButtons.forEach(btn => {
      btn.addEventListener('click', (e) => {
        const tab = e.currentTarget.dataset.tab;
        this.setActiveTab(tab);
      });
    });

    // Close modal
    this.modalCloseBtn.addEventListener('click', () => this.closeModal());
    this.modalOverlay.addEventListener('click', (e) => {
      if (e.target === this.modalOverlay) this.closeModal();
    });

    // Action buttons
    if (this.btnNewStudent) {
      this.btnNewStudent.addEventListener('click', () => this.openAddStudentModal());
    }
    if (this.btnExportData) {
      this.btnExportData.addEventListener('click', () => this.exportDatabaseJSON());
    }
    if (this.btnResetData) {
      this.btnResetData.addEventListener('click', () => {
        if (confirm('Reset institutional database to original IEEE seed records?')) {
          this.dal.resetToDefaults();
          this.showToast('Database reset to IEEE sample dataset', 'info');
          this.render();
        }
      });
    }
  }

  setRole(role) {
    this.currentRole = role;
    this.roleButtons.forEach(btn => {
      btn.classList.toggle('active', btn.dataset.role === role);
    });
    if (this.activeRoleBadge) {
      this.activeRoleBadge.textContent = role.toUpperCase();
    }
```

```
      this.showToast(`Switched workspace role to ${role.toUpperCase()}`, 'info');
      this.render();
    }

    setActiveTab(tab) {
      this.activeTab = tab;
      this.tabButtons.forEach(btn => {
        btn.classList.toggle('active', btn.dataset.tab === tab);
      });
      this.render();
    }

    render() {
      this.renderMetrics();

      switch (this.activeTab) {
        case 'directory':
          this.renderStudentDirectoryView();
          break;
        case 'academics':
          this.renderAcademicsView();
          break;
        case 'placements':
          this.renderPlacementsView();
          break;
        case 'research':
          this.renderResearchView();
          break;
        case 'architecture':
          this.renderArchitectureView();
          break;
        case 'communications':
          this.renderCommunicationsView();
          break;
        default:
          this.renderStudentDirectoryView();
      }
    }

    renderMetrics() {
      const metrics = this.bll.getInstitutionalMetrics();
      if (this.metricTotalStudents) this.metricTotalStudents.textContent = metrics.total;
      if (this.metricAvgCgpa) this.metricAvgCgpa.textContent = metrics.avgCgpa;
      if (this.metricPlacementRate) this.metricPlacementRate.textContent = metrics.placementRate;
      if (this.metricPublications) this.metricPublications.textContent = metrics.publicationsCount;
    }

    // ============================================================================
    // VIEW: STUDENT DIRECTORY (IEEE 10058674 & 10516390)
    // ============================================================================
    renderStudentDirectoryView() {
      const depts = this.dal.getDepartments();
      const students = this.bll.filterStudents();

      this.mainContent.innerHTML = `
        <div class="glass-panel">
          <div class="panel-header">
            <div class="panel-title-group">
              <h2>
                <span>Student Module Master Records</span>
                <span class="badge badge-cyan" id="filtered-count-badge">${students.length} Records</span>
              </h2>
              <p>Comprehensive student information, academic credentials, promotion status, and institutiona
l records.</p>
            </div>
```

```
            <div class="toolbar-controls">
              <div class="search-input-wrapper">
                <svg class="search-icon" width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="curr
entColor" stroke-width="2">
                  <circle cx="11" cy="11" r="8"></circle>
                  <line x1="21" y1="21" x2="16.65" y2="16.65"></line>
                </svg>
                <input type="text" id="student-search-input" placeholder="Search by name, roll no, email..."
/>
              </div>

              <select id="filter-dept-select" class="select-filter">
                <option value="all">All Departments</option>
                ${depts.map(d => `<option value="${d.id}">${d.name} (${d.code})</option>`).join('')}
              </select>

              <select id="filter-status-select" class="select-filter">
                <option value="all">All Statuses</option>
                <option value="Regular">Regular</option>
                <option value="Graduating Senior">Graduating Senior</option>
                <option value="Placed">Placed Candidates</option>
                <option value="Eligible">Eligible for Placements</option>
              </select>

              ${this.currentRole === 'admin' ? `
                <button class="btn-quick btn-primary" id="btn-add-student-inline">
                  <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-w
idth="2.5">
                    <line x1="12" y1="5" x2="12" y2="19"></line>
                    <line x1="5" y1="12" x2="19" y2="12"></line>
                  </svg>
                  Enroll New Student
                </button>
              ` : ''}
            </div>
          </div>

          <div class="table-responsive">
            <table class="data-table">
              <thead>
                <tr>
                  <th>Student Dossier</th>
                  <th>Department & Major</th>
                  <th>Semester / Batch</th>
                  <th>CGPA & Credits</th>
                  <th>Placement Status</th>
                  <th>Promotion Progress</th>
                  <th style="text-align: right;">Actions</th>
                </tr>
              </thead>
              <tbody id="student-table-body">
                ${this.buildStudentTableRows(students)}
              </tbody>
            </table>
          </div>
        </div>
      `;

      // Hook search & filter inputs
      const searchInput = document.getElementById('student-search-input');
      const deptSelect = document.getElementById('filter-dept-select');
      const statusSelect = document.getElementById('filter-status-select');
      const tableBody = document.getElementById('student-table-body');
      const countBadge = document.getElementById('filtered-count-badge');

```

```
    const updateFilterResults = () => {
      const q = searchInput.value;
      const dept = deptSelect.value;
      const st = statusSelect.value;
      const filtered = this.bll.filterStudents(q, dept, st);
      tableBody.innerHTML = this.buildStudentTableRows(filtered);
      countBadge.textContent = `${filtered.length} Records`;
      this.attachTableRowEventListeners();
    };

    searchInput.addEventListener('input', updateFilterResults);
    deptSelect.addEventListener('change', updateFilterResults);
    statusSelect.addEventListener('change', updateFilterResults);

    const inlineAdd = document.getElementById('btn-add-student-inline');
    if (inlineAdd) {
      inlineAdd.addEventListener('click', () => this.openAddStudentModal());
    }

    this.attachTableRowEventListeners();
  }

  buildStudentTableRows(students) {
    if (students.length === 0) {
      return `
        <tr>
          <td colspan="7" style="text-align: center; padding: 2.5rem; color: var(--text-muted);">
            No student records matching your query criteria.
          </td>
        </tr>
      `;
    }

    const depts = this.dal.getDepartments();
    const majors = this.dal.getMajors();

    return students.map(s => {
      const dept = depts.find(d => d.id === s.deptId);
      const major = majors.find(m => m.id === s.majorId);
      const deptCode = dept ? dept.code : 'UNASSIGNED';
      const majorName = major ? major.name : 'General';
      const initials = s.name.split(' ').map(n => n[0]).join('').slice(0, 2).toUpperCase();

      let cgpaClass = 'high';
      if (s.cgpa < 8.0) cgpaClass = 'med';
      if (s.cgpa < 7.0) cgpaClass = 'low';

      let placementBadge = '<span class="badge badge-amber">Unassigned</span>';
      if (s.placement) {
        if (s.placement.status === 'Placed') {
          placementBadge = `<span class="badge badge-emerald">Placed • ${s.placement.company}</span>`;
        } else if (s.placement.status === 'Interviewing') {
          placementBadge = `<span class="badge badge-purple">Interviewing • ${s.placement.company}</span>
`;
        } else if (s.placement.status === 'Eligible') {
          placementBadge = `<span class="badge badge-cyan">Eligible Candidate</span>`;
        }
      }

      return `
        <tr data-student-id="${s.id}">
          <td>
            <div class="student-identity">
              <div class="student-avatar">${initials}</div>
              <div>
```

```
                    <div class="student-name">${s.name}</div>
                    <div class="student-roll">${s.rollNo} • ${s.email}</div>
                  </div>
                </div>
              </td>
              <td>
                <span class="badge badge-cyan">${deptCode}</span>
                <div style="font-size: 0.75rem; color: var(--text-muted); margin-top: 0.2rem;">${majorName}</d
iv>
              </td>
              <td>
                <span style="font-weight: 600;">Sem ${s.currentSemester}</span>
                <div style="font-size: 0.725rem; color: var(--text-dim);">Adm: ${s.admissionYear}</div>
              </td>
              <td>
                <div class="cgpa-indicator">
                  <span class="cgpa-pill ${cgpaClass}">${s.cgpa.toFixed(2)}</span>
                  <span style="font-size: 0.75rem; color: var(--text-muted);">${s.totalCreditsEarned} Credits
</span>
                </div>
              </td>
              <td>${placementBadge}</td>
              <td>
                <div style="font-size: 0.8rem; font-weight: 600; color: #38bdf8;">${s.promotionStatus || 'Acti
ve'}</div>
                <div style="font-size: 0.7rem; color: var(--text-dim);">${s.academicStatus}</div>
              </td>
              <td style="text-align: right;">
                <div class="action-btn-group" style="justify-content: flex-end;">
                  <button class="action-icon-btn btn-view-dossier" title="View Detailed Student Dossier" data-
id="${s.id}">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-w
idth="2">
                      <path d="M1 12s4-8 11-8 11 8 11 8-4 8-11 8-11-8-11-8z"></path>
                      <circle cx="12" cy="12" r="3"></circle>
                    </svg>
                  </button>

                  ${this.currentRole !== 'student' ? `
                    <button class="action-icon-btn btn-edit-student" title="Edit Student Profile" data-id
="${s.id}">
                      <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke
-width="2">
                        <path d="M11 4H4a2 2 0 0 0-2 2v14a2 2 0 0 0 2 2h14a2 2 0 0 0 2-2v-7"></path>
                        <path d="M18.5 2.5a2.121 2.121 0 0 1 3 3L12 15l-4 1 1-4 9.5-9.5z"></path>
                      </svg>
                    </button>
                  ` : ''}

                  ${this.currentRole === 'admin' ? `
                    <button class="action-icon-btn btn-promote-student" title="Promote Student to Next Semeste
r" data-id="${s.id}">
                      <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke
-width="2">
                        <polyline points="18 15 12 9 6 15"></polyline>
                      </svg>
                    </button>
                    <button class="action-icon-btn delete btn-delete-student" title="Purge Record" data-id
="${s.id}">
                      <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke
-width="2">
                        <polyline points="3 6 5 6 21 6"></polyline>
                        <path d="M19 6v14a2 2 0 0 1-2 2H7a2 2 0 0 1-2-2V6m3 0V4a2 2 0 0 1 2-2h4a2 2 0 0 1 2 2v
2"></path>
                      </svg>
```

```
                </button>
              ` : ''}
            </div>
          </td>
        </tr>
      `;
    }).join('');
  }

  attachTableRowEventListeners() {
    // Dossier view
    document.querySelectorAll('.btn-view-dossier').forEach(btn => {
      btn.addEventListener('click', (e) => {
        const id = e.currentTarget.dataset.id;
        this.openStudentDossierModal(id);
      });
    });

    // Edit modal
    document.querySelectorAll('.btn-edit-student').forEach(btn => {
      btn.addEventListener('click', (e) => {
        const id = e.currentTarget.dataset.id;
        this.openEditStudentModal(id);
      });
    });

    // Promote action
    document.querySelectorAll('.btn-promote-student').forEach(btn => {
      btn.addEventListener('click', (e) => {
        const id = e.currentTarget.dataset.id;
        const res = this.bll.promoteStudentSemester(id);
        if (res.success) {
          this.showToast(res.message, 'success');
          this.render();
        }
      });
    });

    // Delete record
    document.querySelectorAll('.btn-delete-student').forEach(btn => {
      btn.addEventListener('click', (e) => {
        const id = e.currentTarget.dataset.id;
        const student = this.dal.getStudentById(id);
        if (confirm(`Are you sure you want to delete student "${student.name}" (${student.rollNo})?`)) {
          this.dal.deleteStudent(id);
          this.showToast(`Deleted student record for ${student.name}`, 'info');
          this.render();
        }
      });
    });
  }

  // ============================================================================
  // VIEW: ACADEMICS & COMPREHENSIVE SCORES (IEEE 7384005)
  // ============================================================================
  renderAcademicsView() {
    const students = this.dal.getStudents();
    const courses = this.dal.getCourses();

    this.mainContent.innerHTML = `
      <div class="glass-panel">
        <div class="panel-header">
          <div class="panel-title-group">
            <h2>
              <span>Comprehensive Score & Grade Management</span>
```

```
                <span class="badge badge-emerald">Tier-2 BLL Active</span>
              </h2>
              <p>Based on IEEE 7384005: 4-Component Score Formula (Internal Assessment + Assignments + Semes
ter Exam = Total Grade Point).</p>
            </div>
            ${this.currentRole === 'faculty' || this.currentRole === 'admin' ? `
              <button class="btn-quick btn-primary" id="btn-grade-entry-trigger">
                <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-wid
th="2">
                  <path d="M12 20h9"></path>
                  <path d="M16.5 3.5a2.121 2.121 0 0 1 3 3L7 19l-4 1 1-4L16.5 3.5z"></path>
                </svg>
                Faculty Grade Entry
              </button>
            ` : ''}
          </div>

          <div class="charts-grid">
            <div class="chart-card">
              <div class="chart-header">
                <span>Grade Distribution Overview</span>
                <span class="badge badge-cyan">Institutional Cohort</span>
              </div>
              <div class="bar-chart-container">
                <div class="bar-row">
                  <div class="bar-row-header">
                    <span>Grade O (Outstanding 90-100)</span>
                    <span style="font-weight: 700; color: #34d399;">52%</span>
                  </div>
                  <div class="bar-track"><div class="bar-fill" style="width: 52%; background: #10b981;"></di
v></div>
                </div>
                <div class="bar-row">
                  <div class="bar-row-header">
                    <span>Grade A+ (Excellent 80-89)</span>
                    <span style="font-weight: 700; color: #38bdf8;">34%</span>
                  </div>
                  <div class="bar-track"><div class="bar-fill" style="width: 34%; background: #06b6d4;"></di
v></div>
                </div>
                <div class="bar-row">
                  <div class="bar-row-header">
                    <span>Grade A (Very Good 70-79)</span>
                    <span style="font-weight: 700; color: #818cf8;">11%</span>
                  </div>
                  <div class="bar-track"><div class="bar-fill" style="width: 11%; background: #6366f1;"></di
v></div>
                </div>
                <div class="bar-row">
                  <div class="bar-row-header">
                    <span>Grade B+ / Below (Remedial)</span>
                    <span style="font-weight: 700; color: #fbbf24;">3%</span>
                  </div>
                  <div class="bar-track"><div class="bar-fill" style="width: 3%; background: #f59e0b;"></div
></div>
                </div>
              </div>
            </div>

            <div class="chart-card">
              <div class="chart-header">
                <span>Active Curriculum Courses</span>
                <span class="badge badge-purple">${courses.length} Modules</span>
              </div>
              <div style="display: flex; flex-direction: column; gap: 0.6rem;">
```

```
                ${courses.map(c => `
                  <div style="background: var(--bg-secondary); border: 1px solid var(--border-subtle); paddi
ng: 0.65rem 0.85rem; border-radius: var(--radius-sm); display: flex; justify-content: space-between; align-items: c
enter;">
                    <div>
                      <div style="font-size: 0.825rem; font-weight: 600;">${c.code}: ${c.name}</div>
                      <div style="font-size: 0.7rem; color: var(--text-muted);">${c.type} • ${c.credits} Cre
dits</div>
                    </div>
                    <span class="badge badge-cyan">${c.dept}</span>
                  </div>
                `).join('')}
              </div>
            </div>
          </div>

          <div class="table-responsive" style="margin-top: 1.5rem;">
            <table class="data-table">
              <thead>
                <tr>
                  <th>Student</th>
                  <th>Course Evaluation</th>
                  <th>Internal (30)</th>
                  <th>Assignments (20)</th>
                  <th>Final Exam (50)</th>
                  <th>Total (100)</th>
                  <th>Grade Letter</th>
                  <th>Grade Point</th>
                </tr>
              </thead>
              <tbody>
                ${students.flatMap(s => (s.scores || []).map(score => `
                  <tr>
                    <td>
                      <div style="font-weight: 600;">${s.name}</div>
                      <div style="font-size: 0.725rem; color: var(--text-muted);">${s.rollNo}</div>
                    </td>
                    <td>
                      <span class="badge badge-purple">${score.courseCode}</span>
                      <span style="font-size: 0.8rem; margin-left: 0.3rem;">${score.courseName}</span>
                    </td>
                    <td>${score.internal} / 30</td>
                    <td>${score.assignments} / 20</td>
                    <td>${score.exam} / 50</td>
                    <td style="font-weight: 700; color: #fff;">${score.total}</td>
                    <td><span class="badge badge-emerald">${score.grade}</span></td>
                    <td><span class="cgpa-pill high">${(score.gpa || 0).toFixed(1)}</span></td>
                  </tr>
                `)).join('')}
              </tbody>
            </table>
          </div>
        </div>
      `;

      const gradeTrigger = document.getElementById('btn-grade-entry-trigger');
      if (gradeTrigger) {
        gradeTrigger.addEventListener('click', () => this.openFacultyGradeModal());
      }
    }

    // ============================================================================
    // VIEW: PLACEMENT & INTERNSHIP DIRECTORATE (IEEE 10058674)
    // ============================================================================
    renderPlacementsView() {
```

```
744       const students = this.dal.getStudents();
745       const placedStudents = students.filter(s => s.placement && s.placement.status === 'Placed');
746
747       this.mainContent.innerHTML = `
748         <div class="glass-panel">
749           <div class="panel-header">
750             <div class="panel-title-group">
751               <h2>
752                 <span>Corporate Placement & Internship Directorate</span>
753                 <span class="badge badge-emerald">Campus Drive 2025-2026</span>
754               </h2>
755               <p>From IEEE 10058674: Student placement training records, interviews, offers, and compensatio
n CTC tracking.</p>
756             </div>
757             <button class="btn-quick btn-primary" id="btn-check-eligibility">
758               <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width
="2">
759                 <polyline points="9 11 12 14 22 4"></polyline>
760                 <path d="M21 12v7a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2V5a2 2 0 0 1 2-2h11"></path>
761               </svg>
762               Run Eligibility Audit
763             </button>
764           </div>
765
766           <div class="charts-grid">
767             <div class="chart-card">
768               <div class="chart-header">
769                 <span>Top Recruiting Partners</span>
770                 <span class="badge badge-emerald">Tier-1 Marquee</span>
771               </div>
772               <div style="display: flex; flex-direction: column; gap: 0.75rem;">
773                 <div style="display: flex; justify-content: space-between; align-items: center; background:
var(--bg-secondary); padding: 0.75rem 1rem; border-radius: var(--radius-sm); border: 1px solid var(--border-subtl
e);">
774                   <div>
775                     <div style="font-weight: 700; color: #fff;">NVIDIA AI Labs</div>
776                     <div style="font-size: 0.725rem; color: var(--text-muted);">Systems Software & Parallel
Computing</div>
777                   </div>
778                   <div style="text-align: right;">
779                     <div style="font-weight: 700; color: #34d399;">32.5 LPA</div>
780                     <span class="badge badge-emerald">1 Offer</span>
781                   </div>
782                 </div>
783
784                 <div style="display: flex; justify-content: space-between; align-items: center; background:
var(--bg-secondary); padding: 0.75rem 1rem; border-radius: var(--radius-sm); border: 1px solid var(--border-subtl
e);">
785                   <div>
786                     <div style="font-weight: 700; color: #fff;">Google DeepMind</div>
787                     <div style="font-size: 0.725rem; color: var(--text-muted);">Research Associate Internshi
p</div>
788                   </div>
789                   <div style="text-align: right;">
790                     <div style="font-weight: 700; color: #38bdf8;">38.0 LPA</div>
791                     <span class="badge badge-cyan">1 Offer</span>
792                   </div>
793                 </div>
794
795                 <div style="display: flex; justify-content: space-between; align-items: center; background:
var(--bg-secondary); padding: 0.75rem 1rem; border-radius: var(--radius-sm); border: 1px solid var(--border-subtl
e);">
796                   <div>
797                     <div style="font-weight: 700; color: #fff;">Microsoft Azure Core</div>
798                     <div style="font-size: 0.725rem; color: var(--text-muted);">Cloud Infrastructure Enginee
```

```
ring</div>
                  </div>
                  <div style="text-align: right;">
                    <div style="font-weight: 700; color: #818cf8;">28.0 LPA</div>
                    <span class="badge badge-purple">1 Offer</span>
                  </div>
                </div>
              </div>
            </div>

            <div class="chart-card">
              <div class="chart-header">
                <span>Placement Status Ratio</span>
                <span class="badge badge-cyan">Active Batch</span>
              </div>
              <div class="bar-chart-container">
                <div class="bar-row">
                  <div class="bar-row-header">
                    <span>Placed Candidates</span>
                    <span style="font-weight: 700; color: #34d399;">60%</span>
                  </div>
                  <div class="bar-track"><div class="bar-fill" style="width: 60%; background: #10b981;"></di
v></div>
                </div>
                <div class="bar-row">
                  <div class="bar-row-header">
                    <span>Final Round Interviewing</span>
                    <span style="font-weight: 700; color: #38bdf8;">20%</span>
                  </div>
                  <div class="bar-track"><div class="bar-fill" style="width: 20%; background: #06b6d4;"></di
v></div>
                </div>
                <div class="bar-row">
                  <div class="bar-row-header">
                    <span>Eligible for Upcoming Drives</span>
                    <span style="font-weight: 700; color: #fbbf24;">20%</span>
                  </div>
                  <div class="bar-track"><div class="bar-fill" style="width: 20%; background: #f59e0b;"></di
v></div>
                </div>
              </div>
            </div>
          </div>

          <div class="table-responsive">
            <table class="data-table">
              <thead>
                <tr>
                  <th>Candidate</th>
                  <th>Roll No</th>
                  <th>CGPA</th>
                  <th>Company</th>
                  <th>Role Designation</th>
                  <th>Compensation (CTC)</th>
                  <th>Offer Date</th>
                  <th>Eligibility Check</th>
                </tr>
              </thead>
              <tbody>
                ${students.map(s => {
                  const p = s.placement || { status: 'Not registered', company: '—', role: '—', ctc: '—', of
ferDate: '—' };
                  const elig = this.bll.checkPlacementEligibility(s);
                  return `
                    <tr>
```

```
                        <td style="font-weight: 600;">${s.name}</td>
                        <td style="font-family: var(--font-mono); font-size: 0.75rem;">${s.rollNo}</td>
                        <td><span class="cgpa-pill ${s.cgpa >= 8.5 ? 'high' : 'med'}">${s.cgpa.toFixed(2)}</span></td>
                        <td style="font-weight: 600; color: #38bdf8;">${p.company}</td>
                        <td>${p.role}</td>
                        <td style="font-weight: 700; color: #34d399;">${p.ctc}</td>
                        <td style="font-size: 0.75rem; color: var(--text-muted);">${p.offerDate}</td>
                        <td>
                          <span class="badge ${elig.eligible ? 'badge-emerald' : 'badge-amber'}">
                            ${elig.statusText}
                          </span>
                        </td>
                      </tr>
                    `;
                  }).join('')}
                </tbody>
              </table>
            </div>
          </div>
        `;

        const auditBtn = document.getElementById('btn-check-eligibility');
        if (auditBtn) {
          auditBtn.addEventListener('click', () => {
            const eligibleCount = students.filter(s => this.bll.checkPlacementEligibility(s).eligible).length;
            this.showToast(`Eligibility Audit Complete: ${eligibleCount} of ${students.length} students satisfy Tier-1 criteria!`, 'success');
          });
        }
      }

      // ============================================================================
      // VIEW: RESEARCH, CAPSTONES & EVENTS (IEEE 10058674)
      // ============================================================================
      renderResearchView() {
        const students = this.dal.getStudents();

        this.mainContent.innerHTML = `
          <div class="glass-panel">
            <div class="panel-header">
              <div class="panel-title-group">
                <h2>
                  <span>Research, Publications & Student Capstones</span>
                  <span class="badge badge-purple">Academic Output</span>
                </h2>
                <p>From IEEE 10058674: Module for tracking student publishing papers/journals, patent applications, project guides, and symposiums.</p>
              </div>
            </div>

            <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(340px, 1fr)); gap: 1.25rem;">
              ${students.flatMap(s => (s.projectsAndPublications || []).map(p => `
                <div class="item-chip">
                  <div class="chip-head">
                    <span class="badge ${p.type.includes('Paper') ? 'badge-cyan' : 'badge-emerald'}">${p.type}</span>
                    <span style="font-size: 0.725rem; color: var(--text-muted);">${s.name} (${s.rollNo})</span>
                  </div>
                  <div class="chip-title" style="margin: 0.5rem 0;">${p.title}</div>
                  <div class="chip-sub">
                    <strong>Target Venue:</strong> ${p.venue}
                  </div>
```

```
                <div class="chip-sub" style="margin-top: 0.25rem;">
                  <strong>Faculty Supervisor:</strong> ${p.guide}
                </div>
              </div>
            `)).join('')}
          </div>

          <h3 style="margin: 2rem 0 1rem; font-size: 1.15rem; color: var(--text-main); display: flex; align-items: center; gap: 0.5rem;">
            <span>Student Training, Competitions & Events Attended</span>
            <span class="badge badge-cyan">IEEE 10058674 Activity Track</span>
          </h3>

          <div class="table-responsive">
            <table class="data-table">
              <thead>
                <tr>
                  <th>Participant Student</th>
                  <th>Event / Symposium Name</th>
                  <th>Activity Category</th>
                  <th>Outcome / Honors Achieved</th>
                </tr>
              </thead>
              <tbody>
                ${students.flatMap(s => (s.events || []).map(e => `
                  <tr>
                    <td style="font-weight: 600;">${s.name} (${s.rollNo})</td>
                    <td style="font-weight: 600; color: #fff;">${e.name}</td>
                    <td><span class="badge badge-purple">${e.type}</span></td>
                    <td><span class="badge badge-emerald">${e.outcome}</span></td>
                  </tr>
                `)).join('')}
              </tbody>
            </table>
          </div>
        </div>
      `;
    }

    // ================================================================================
    // VIEW: 4-TIER ARCHITECTURE & ER SCHEMA (IEEE 7384005)
    // ================================================================================
    renderArchitectureView() {
      this.mainContent.innerHTML = `
        <div class="glass-panel">
          <div class="panel-header">
            <div class="panel-title-group">
              <h2>
                <span>Hierarchical 4-Tier Computer-Aided System Architecture</span>
                <span class="badge badge-cyan">IEEE 7384005 Specifications</span>
              </h2>
              <p>Direct architectural mapping of the computer-aided college student management system proposed in IEEE 7384005.</p>
            </div>
          </div>

          <div class="architecture-grid">
            <div class="tier-column active-tier">
              <div class="tier-header">
                <div class="tier-tag">Layer 1</div>
                <div class="tier-title">Web Display Layer</div>
              </div>
              <div class="tier-component-list">
                <div class="tier-box">
                  <div class="box-title">Student Portal View <span class="badge badge-cyan">Active</span></d
```

```
iv>
                  <div class="box-desc">Transcript queries, enrolled courses, placement status, and publicat
ion showcase.</div>
                </div>
                <div class="tier-box">
                  <div class="box-title">Teacher / Faculty View</div>
                  <div class="box-desc">Grade recording, attendance audit, and capstone supervision workflow
s.</div>
                </div>
                <div class="tier-box">
                  <div class="box-title">Manager / Admin View</div>
                  <div class="box-desc">Student onboarding, promotion engines, department configuration, and
institutional metrics.</div>
                </div>
              </div>
            </div>

            <div class="tier-column">
              <div class="tier-header">
                <div class="tier-tag">Layer 2</div>
                <div class="tier-title">Business Logic Layer</div>
              </div>
              <div class="tier-component-list">
                <div class="tier-box">
                  <div class="box-title">Score & GPA Calculator</div>
                  <div class="box-desc">4-component evaluation: Internal + Assignments + Final Exam = Compre
hensive Grade.</div>
                </div>
                <div class="tier-box">
                  <div class="box-title">Promotion Engine</div>
                  <div class="box-desc">Automated validation of credits, prerequisites, and semester progres
sion.</div>
                </div>
                <div class="tier-box">
                  <div class="box-title">Placement Eligibility</div>
                  <div class="box-desc">CGPA thresholds and backlog checks for corporate drive shortlisting.
</div>
                </div>
              </div>
            </div>

            <div class="tier-column">
              <div class="tier-header">
                <div class="tier-tag">Layer 3</div>
                <div class="tier-title">Data Access Layer (DAL)</div>
              </div>
              <div class="tier-component-list">
                <div class="tier-box">
                  <div class="box-title">Student Repository</div>
                  <div class="box-desc">Normalized persistence, multi-predicate filtering, and transactional
updates.</div>
                </div>
                <div class="tier-box">
                  <div class="box-title">Schema Relational Mapper</div>
                  <div class="box-desc">Maintains foreign key integrity across departments, majors, classes,
and teachers.</div>
                </div>
                <div class="tier-box">
                  <div class="box-title">Data Serialization Engine</div>
                  <div class="box-desc">JSON backup/restore, export pipelines, and caching layers.</div>
                </div>
              </div>
            </div>

            <div class="tier-column">
```

```
              <div class="tier-header">
                <div class="tier-tag">Layer 4</div>
                <div class="tier-title">Database Layer</div>
              </div>
              <div class="tier-component-list">
                <div class="tier-box">
                  <div class="box-title">Central Storage Schema</div>
                  <div class="box-desc">Consolidated tables with zero redundancy as highlighted in IEEE 1005
8674.</div>
                </div>
                <div class="tier-box">
                  <div class="box-title">Audit Log Store</div>
                  <div class="box-desc">Historical promotion logs, transfer histories, and grade alteration
records.</div>
                </div>
                <div class="tier-box">
                  <div class="box-title">Communication DB</div>
                  <div class="box-desc">Institutional broadcasts and notice boards (IEEE 10516390).</div>
                </div>
              </div>
            </div>
          </div>

          <h3 style="margin: 2.5rem 0 1rem; font-size: 1.15rem; color: var(--text-main); display: flex; alig
n-items: center; gap: 0.5rem;">
            <span>Entity-Relationship (ER) Schema Specifications</span>
            <span class="badge badge-emerald">IEEE 7384005 ER Design</span>
          </h3>
          <p style="font-size: 0.8rem; color: var(--text-muted); margin-bottom: 1rem;">
            Relational model defining entities, Primary Keys (PK), and Foreign Keys (FK) connecting student
s, departments, majors, classes, courses, and scores.
          </p>

          <div class="er-container">
            <!-- STUDENT ENTITY -->
            <div class="er-card">
              <div class="er-card-header">
                <span>STUDENT_INFO</span>
                <span class="badge badge-cyan">Entity</span>
              </div>
              <div class="er-card-body">
                <div class="er-field"><span class="er-field-name">id</span><div class="er-field-meta"><span
class="pk-tag">PK</span><span class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">rollNo</span><div class="er-field-meta"><s
pan class="er-field-type">VARCHAR UNIQUE</span></div></div>
                <div class="er-field"><span class="er-field-name">name</span><div class="er-field-meta"><spa
n class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">email</span><div class="er-field-meta"><sp
an class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">deptId</span><div class="er-field-meta"><s
pan class="fk-tag">FK &rarr; DEPT</span><span class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">majorId</span><div class="er-field-meta"><
span class="fk-tag">FK &rarr; MAJOR</span><span class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">classId</span><div class="er-field-meta"><
span class="fk-tag">FK &rarr; CLASS</span><span class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">cgpa</span><div class="er-field-meta"><spa
n class="er-field-type">DECIMAL(4,2)</span></div></div>
                <div class="er-field"><span class="er-field-name">promotionStatus</span><div class="er-field
-meta"><span class="er-field-type">VARCHAR</span></div></div>
              </div>
            </div>

            <!-- TEACHER ENTITY -->
            <div class="er-card">
              <div class="er-card-header">
```

```
                <span>TEACHER_INFO</span>
                <span class="badge badge-emerald">Entity</span>
              </div>
              <div class="er-card-body">
                <div class="er-field"><span class="er-field-name">id</span><div class="er-field-meta"><span class="pk-tag">PK</span><span class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">name</span><div class="er-field-meta"><span class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">department</span><div class="er-field-meta"><span class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">title</span><div class="er-field-meta"><span class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">email</span><div class="er-field-meta"><span class="er-field-type">VARCHAR</span></div></div>
              </div>
            </div>

            <!-- DEPARTMENT & MAJOR ENTITY -->
            <div class="er-card">
              <div class="er-card-header">
                <span>DEPARTMENT & MAJOR</span>
                <span class="badge badge-purple">Entity</span>
              </div>
              <div class="er-card-body">
                <div class="er-field"><span class="er-field-name">id</span><div class="er-field-meta"><span class="pk-tag">PK</span><span class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">code</span><div class="er-field-meta"><span class="er-field-type">VARCHAR(10)</span></div></div>
                <div class="er-field"><span class="er-field-name">name</span><div class="er-field-meta"><span class="er-field-type">VARCHAR(100)</span></div></div>
                <div class="er-field"><span class="er-field-name">deptId</span><div class="er-field-meta"><span class="fk-tag">FK &rarr; DEPT</span><span class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">degree</span><div class="er-field-meta"><span class="er-field-type">VARCHAR</span></div></div>
              </div>
            </div>

            <!-- COMPREHENSIVE SCORE ENTITY -->
            <div class="er-card">
              <div class="er-card-header">
                <span>COMPREHENSIVE_SCORE</span>
                <span class="badge badge-amber">Entity</span>
              </div>
              <div class="er-card-body">
                <div class="er-field"><span class="er-field-name">scoreId</span><div class="er-field-meta"><span class="pk-tag">PK</span><span class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">studentId</span><div class="er-field-meta"><span class="fk-tag">FK &rarr; STUDENT</span><span class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">courseCode</span><div class="er-field-meta"><span class="fk-tag">FK &rarr; COURSE</span><span class="er-field-type">VARCHAR</span></div></div>
                <div class="er-field"><span class="er-field-name">internalScore</span><div class="er-field-meta"><span class="er-field-type">INT(30)</span></div></div>
                <div class="er-field"><span class="er-field-name">assignmentScore</span><div class="er-field-meta"><span class="er-field-type">INT(20)</span></div></div>
                <div class="er-field"><span class="er-field-name">examScore</span><div class="er-field-meta"><span class="er-field-type">INT(50)</span></div></div>
                <div class="er-field"><span class="er-field-name">totalScore</span><div class="er-field-meta"><span class="er-field-type">INT(100)</span></div></div>
                <div class="er-field"><span class="er-field-name">gradeLetter</span><div class="er-field-meta"><span class="er-field-type">CHAR(2)</span></div></div>
              </div>
            </div>
          </div>
        </div>
      `;
```

```
    }

    // ============================================================================
    // VIEW: INSTITUTIONAL COMMUNICATIONS (IEEE 10516390)
    // ============================================================================
    renderCommunicationsView() {
      const announcements = this.dal.getAnnouncements();

      this.mainContent.innerHTML = `
        <div class="glass-panel">
          <div class="panel-header">
            <div class="panel-title-group">
              <h2>
                <span>Institutional Broadcast & Communication Hub</span>
                <span class="badge badge-cyan">IEEE 10516390 Feature</span>
              </h2>
              <p>Direct communication channel connecting Administrators, Instructors, and Students to stream
line institution operations.</p>
            </div>
            ${this.currentRole === 'admin' ? `
              <button class="btn-quick btn-primary" id="btn-post-announcement">
                <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-wid
th="2">
                  <line x1="12" y1="5" x2="12" y2="19"></line>
                  <line x1="5" y1="12" x2="19" y2="12"></line>
                </svg>
                Publish New Notice
              </button>
            ` : ''}
          </div>

          <div style="display: flex; flex-direction: column; gap: 1rem;">
            ${announcements.map(a => `
              <div style="background: var(--bg-secondary); border: 1px solid var(--border-subtle); border-ra
dius: var(--radius-md); padding: 1.25rem;">
                <div style="display: flex; justify-content: space-between; align-items: center; margin-botto
m: 0.5rem; flex-wrap: wrap; gap: 0.5rem;">
                  <div style="display: flex; align-items: center; gap: 0.5rem;">
                    <span class="badge ${a.priority === 'High' ? 'badge-rose' : (a.priority === 'Important'
? 'badge-amber' : 'badge-cyan')}">${a.priority} Priority</span>
                    <h3 style="font-size: 1rem; font-weight: 700; color: #fff;">${a.title}</h3>
                  </div>
                  <div style="font-size: 0.75rem; color: var(--text-muted);">
                    Published: <strong>${a.date}</strong> by <em>${a.author}</em>
                  </div>
                </div>
                <p style="font-size: 0.85rem; color: var(--text-muted); line-height: 1.6;">${a.content}</p>
              </div>
            `).join('')}
          </div>
        </div>
      `;

      const postBtn = document.getElementById('btn-post-announcement');
      if (postBtn) {
        postBtn.addEventListener('click', () => this.openPostAnnouncementModal());
      }
    }

    // ============================================================================
    // MODALS & INTERACTIVE WORKFLOWS
    // ============================================================================
    openModal(title, bodyHtml, footerHtml) {
      this.modalTitle.textContent = title;
      this.modalBody.innerHTML = bodyHtml;
```

```
      this.modalFooter.innerHTML = footerHtml;
      this.modalOverlay.classList.add('active');
    }

    closeModal() {
      this.modalOverlay.classList.remove('active');
    }

    openStudentDossierModal(studentId) {
      const s = this.dal.getStudentById(studentId);
      if (!s) return;

      const dept = this.dal.getDepartments().find(d => d.id === s.deptId);
      const major = this.dal.getMajors().find(m => m.id === s.majorId);
      const cls = this.dal.getClasses().find(c => c.id === s.classId);
      const initials = s.name.split(' ').map(n => n[0]).join('').slice(0, 2).toUpperCase();

      const bodyHtml = `
        <div class="dossier-grid">
          <div class="dossier-sidebar">
            <div class="dossier-avatar-large">${initials}</div>
            <h4 style="font-size: 1.1rem; font-weight: 700; color: #fff;">${s.name}</h4>
            <span class="badge badge-cyan" style="margin-top: 0.25rem;">${s.rollNo}</span>
            <div style="font-size: 0.75rem; color: var(--text-muted); margin-top: 0.4rem;">${s.email}</div>

            <div class="dossier-info-list">
              <div class="dossier-info-item"><span class="k">Department</span><span class="v">${dept ? dept.name : 'CSE'}</span></div>
              <div class="dossier-info-item"><span class="k">Major Degree</span><span class="v">${major ? major.name : 'B.Tech'}</span></div>
              <div class="dossier-info-item"><span class="k">Cohort / Class</span><span class="v">${cls ? cls.name : 'Alpha'}</span></div>
              <div class="dossier-info-item"><span class="k">Current Sem</span><span class="v">Semester ${s.currentSemester}</span></div>
              <div class="dossier-info-item"><span class="k">Cumulative CGPA</span><span class="v" style="color: #34d399; font-weight: 700;">${s.cgpa.toFixed(2)}</span></div>
              <div class="dossier-info-item"><span class="k">Credits Earned</span><span class="v">${s.totalCreditsEarned} Credits</span></div>
              <div class="dossier-info-item"><span class="k">Admission Year</span><span class="v">${s.admissionYear}</span></div>
            </div>
          </div>

          <div class="dossier-main">
            <!-- Comprehensive Scores -->
            <div class="dossier-section-title">
              <span>Academic Performance & Course Scores</span>
              <span class="badge badge-emerald">IEEE 7384005</span>
            </div>
            <div class="table-responsive" style="margin-bottom: 1.5rem;">
              <table class="data-table">
                <thead>
                  <tr>
                    <th>Course</th>
                    <th>Internal</th>
                    <th>Assignments</th>
                    <th>Exam</th>
                    <th>Total</th>
                    <th>Grade</th>
                  </tr>
                </thead>
                <tbody>
                  ${(s.scores || []).map(score => `
                    <tr>
                      <td><strong>${score.courseCode}</strong>: ${score.courseName}</td>
```

```
                        <td>${score.internal}/30</td>
                        <td>${score.assignments}/20</td>
                        <td>${score.exam}/50</td>
                        <td style="font-weight: 700; color: #fff;">${score.total}</td>
                        <td><span class="badge badge-emerald">${score.grade} (${score.gpa})</span></td>
                      </tr>
                    `).join('')}
                  </tbody>
                </table>
              </div>

              <!-- Placement Record -->
              <div class="dossier-section-title">
                <span>Corporate Placement & Internship Data</span>
                <span class="badge badge-cyan">IEEE 10058674</span>
              </div>
              <div style="background: var(--bg-secondary); border: 1px solid var(--border-subtle); padding: 1rem; border-radius: var(--radius-md); margin-bottom: 1.5rem;">
                ${s.placement ? `
                  <div style="display: flex; justify-content: space-between; align-items: center; flex-wrap: wrap; gap: 0.5rem;">
                    <div>
                      <div style="font-size: 1rem; font-weight: 700; color: #38bdf8;">${s.placement.company}</div>
                      <div style="font-size: 0.8rem; color: var(--text-muted);">${s.placement.role}</div>
                    </div>
                    <div style="text-align: right;">
                      <div style="font-size: 1.1rem; font-weight: 800; color: #34d399;">${s.placement.ctc}</div>
                      <span class="badge badge-emerald">${s.placement.status}</span>
                    </div>
                  </div>
                ` : '<div style="color: var(--text-muted); font-size: 0.8rem;">No corporate placement record linked.</div>'}
              </div>

              <!-- Projects & Publications -->
              <div class="dossier-section-title">
                <span>Published Papers & Capstone Projects</span>
                <span class="badge badge-purple">Research Output</span>
              </div>
              <div style="display: flex; flex-direction: column; gap: 0.5rem; margin-bottom: 1.5rem;">
                ${(s.projectsAndPublications || []).map(p => `
                  <div class="item-chip">
                    <div class="chip-head">
                      <span class="badge badge-purple">${p.type}</span>
                      <span style="font-size: 0.75rem; color: var(--text-muted);">Guide: ${p.guide}</span>
                    </div>
                    <div class="chip-title">${p.title}</div>
                    <div class="chip-sub">Venue: ${p.venue}</div>
                  </div>
                `).join('')}
              </div>

              <!-- Promotion & Transfer History -->
              <div class="dossier-section-title">
                <span>Admission, Transfer & Promotion Audit History</span>
              </div>
              <div style="background: var(--bg-secondary); border: 1px solid var(--border-subtle); padding: 0.85rem; border-radius: var(--radius-md); font-size: 0.8rem; color: var(--text-muted);">
                <div><strong>Promotion Status:</strong> ${s.promotionStatus || 'Normal progression'}</div>
                <div style="margin-top: 0.25rem;"><strong>Transfer / Admission Route:</strong> ${s.transferHistory || 'Direct Regular Enrollment'}</div>
              </div>
            </div>
```

```
        </div>
      `;

      const footerHtml = `
        <button class="btn-quick" id="btn-modal-close-action">Close Dossier</button>
        ${this.currentRole === 'admin' ? `
          <button class="btn-quick btn-primary" id="btn-modal-promote-direct">Promote to Next Semester</button>
        ` : ''}
      `;

      this.openModal(`Official Dossier: ${s.name} (${s.rollNo})`, bodyHtml, footerHtml);

      document.getElementById('btn-modal-close-action').addEventListener('click', () => this.closeModal());
      const promoteDirect = document.getElementById('btn-modal-promote-direct');
      if (promoteDirect) {
        promoteDirect.addEventListener('click', () => {
          const res = this.bll.promoteStudentSemester(s.id);
          if (res.success) {
            this.showToast(res.message, 'success');
            this.closeModal();
            this.render();
          }
        });
      }
    }

    openAddStudentModal() {
      const depts = this.dal.getDepartments();
      const majors = this.dal.getMajors();
      const classes = this.dal.getClasses();

      const bodyHtml = `
        <form id="add-student-form" class="form-grid">
          <div class="form-group">
            <label class="form-label">Full Student Name *</label>
            <input type="text" class="form-control" name="name" required placeholder="e.g. Samantha Wu" />
          </div>
          <div class="form-group">
            <label class="form-label">Roll Number *</label>
            <input type="text" class="form-control" name="rollNo" required placeholder="e.g. 2024CSE108" />
          </div>
          <div class="form-group">
            <label class="form-label">Official University Email *</label>
            <input type="email" class="form-control" name="email" required placeholder="e.g. s.wu@student.ims.edu" />
          </div>
          <div class="form-group">
            <label class="form-label">Department *</label>
            <select class="form-control" name="deptId" required>
              ${depts.map(d => `<option value="${d.id}">${d.name} (${d.code})</option>`).join('')}
            </select>
          </div>
          <div class="form-group">
            <label class="form-label">Major Degree *</label>
            <select class="form-control" name="majorId" required>
              ${majors.map(m => `<option value="${m.id}">${m.name}</option>`).join('')}
            </select>
          </div>
          <div class="form-group">
            <label class="form-label">Cohort / Class Section *</label>
            <select class="form-control" name="classId" required>
              ${classes.map(c => `<option value="${c.id}">${c.name}</option>`).join('')}
            </select>
          </div>
```

```
          <div class="form-group">
            <label class="form-label">Admission Year</label>
            <input type="number" class="form-control" name="admissionYear" value="2024" />
          </div>
          <div class="form-group">
            <label class="form-label">Current Semester</label>
            <input type="number" class="form-control" name="currentSemester" value="4" min="1" max="8" />
          </div>
          <div class="form-group">
            <label class="form-label">Starting CGPA</label>
            <input type="number" step="0.01" class="form-control" name="cgpa" value="8.50" min="0" max="10"
/>
          </div>
          <div class="form-group">
            <label class="form-label">Earned Credits</label>
            <input type="number" class="form-control" name="totalCreditsEarned" value="80" />
          </div>
          <div class="form-group col-span-2">
            <label class="form-label">Admission Route / Transfer History</label>
            <input type="text" class="form-control" name="transferHistory" placeholder="Direct Admission via
State Entrance Exam" />
          </div>
        </form>
      `;

      const footerHtml = `
        <button class="btn-quick" id="btn-cancel-modal">Cancel</button>
        <button class="btn-quick btn-primary" id="btn-submit-student">Save & Enroll Student</button>
      `;

      this.openModal('Enroll New Student Record (IEEE 10058674)', bodyHtml, footerHtml);

      document.getElementById('btn-cancel-modal').addEventListener('click', () => this.closeModal());
      document.getElementById('btn-submit-student').addEventListener('click', () => {
        const form = document.getElementById('add-student-form');
        if (!form.checkValidity()) {
          form.reportValidity();
          return;
        }
        const fd = new FormData(form);
        const newStudent = {
          name: fd.get('name'),
          rollNo: fd.get('rollNo'),
          email: fd.get('email'),
          deptId: fd.get('deptId'),
          majorId: fd.get('majorId'),
          classId: fd.get('classId'),
          admissionYear: parseInt(fd.get('admissionYear'), 10),
          currentSemester: parseInt(fd.get('currentSemester'), 10),
          cgpa: parseFloat(fd.get('cgpa')),
          totalCreditsEarned: parseInt(fd.get('totalCreditsEarned'), 10),
          academicStatus: 'Regular',
          promotionStatus: `Promoted to Sem ${fd.get('currentSemester')}`,
          transferHistory: fd.get('transferHistory') || 'Direct Regular Admission',
          scores: [
            { courseCode: 'CS301', courseName: 'Distributed Systems', internal: 26, assignments: 18, exam: 4
4, total: 88, grade: 'A+', gpa: 9.0 }
          ],
          placement: {
            status: 'Eligible',
            company: 'Campus Placement Cell',
            role: 'Undergraduate Trainee',
            ctc: 'Awaiting Drive Registration',
            offerDate: '—'
          },
```

```
          events: [],
          projectsAndPublications: []
        };

        this.dal.createStudent(newStudent);
        this.showToast(`Enrolled student ${newStudent.name} successfully!`, 'success');
        this.closeModal();
        this.render();
      });
    }

    openEditStudentModal(studentId) {
      const s = this.dal.getStudentById(studentId);
      if (!s) return;

      const depts = this.dal.getDepartments();
      const majors = this.dal.getMajors();
      const classes = this.dal.getClasses();

      const bodyHtml = `
        <form id="edit-student-form" class="form-grid">
          <div class="form-group">
            <label class="form-label">Full Student Name</label>
            <input type="text" class="form-control" name="name" value="${s.name}" required />
          </div>
          <div class="form-group">
            <label class="form-label">Roll Number</label>
            <input type="text" class="form-control" name="rollNo" value="${s.rollNo}" required />
          </div>
          <div class="form-group">
            <label class="form-label">Email</label>
            <input type="email" class="form-control" name="email" value="${s.email}" required />
          </div>
          <div class="form-group">
            <label class="form-label">Department</label>
            <select class="form-control" name="deptId">
              ${depts.map(d => `<option value="${d.id}" ${d.id === s.deptId ? 'selected' : ''}>${d.name}
(${d.code})</option>`).join('')}
            </select>
          </div>
          <div class="form-group">
            <label class="form-label">Current Semester</label>
            <input type="number" class="form-control" name="currentSemester" value="${s.currentSemester}" mi
n="1" max="8" />
          </div>
          <div class="form-group">
            <label class="form-label">Cumulative CGPA</label>
            <input type="number" step="0.01" class="form-control" name="cgpa" value="${s.cgpa}" min="0" max
="10" />
          </div>
          <div class="form-group">
            <label class="form-label">Placement Company</label>
            <input type="text" class="form-control" name="company" value="${s.placement ? s.placement.compan
y : ''}" />
          </div>
          <div class="form-group">
            <label class="form-label">Placement CTC Package</label>
            <input type="text" class="form-control" name="ctc" value="${s.placement ? s.placement.ctc : ''}"
/>
          </div>
          <div class="form-group col-span-2">
            <label class="form-label">Promotion / Status Notes</label>
            <input type="text" class="form-control" name="promotionStatus" value="${s.promotionStatus ||
''}" />
          </div>
```

```
        </form>
      `;

      const footerHtml = `
        <button class="btn-quick" id="btn-cancel-edit">Cancel</button>
        <button class="btn-quick btn-primary" id="btn-save-edit">Update Dossier</button>
      `;

      this.openModal(`Edit Dossier: ${s.name}`, bodyHtml, footerHtml);

      document.getElementById('btn-cancel-edit').addEventListener('click', () => this.closeModal());
      document.getElementById('btn-save-edit').addEventListener('click', () => {
        const form = document.getElementById('edit-student-form');
        const fd = new FormData(form);

        const updated = {
          name: fd.get('name'),
          rollNo: fd.get('rollNo'),
          email: fd.get('email'),
          deptId: fd.get('deptId'),
          currentSemester: parseInt(fd.get('currentSemester'), 10),
          cgpa: parseFloat(fd.get('cgpa')),
          promotionStatus: fd.get('promotionStatus'),
          placement: {
            ...(s.placement || {}),
            company: fd.get('company'),
            ctc: fd.get('ctc')
          }
        };

        this.dal.updateStudent(s.id, updated);
        this.showToast(`Updated profile for ${updated.name}`, 'success');
        this.closeModal();
        this.render();
      });
    }

    openFacultyGradeModal() {
      const students = this.dal.getStudents();
      const courses = this.dal.getCourses();

      const bodyHtml = `
        <form id="grade-entry-form" class="form-grid">
          <div class="form-group">
            <label class="form-label">Select Student *</label>
            <select class="form-control" name="studentId" required>
              ${students.map(s => `<option value="${s.id}">${s.name} (${s.rollNo})</option>`).join('')}
            </select>
          </div>
          <div class="form-group">
            <label class="form-label">Select Course Module *</label>
            <select class="form-control" name="courseCode" required>
              ${courses.map(c => `<option value="${c.code}">${c.code} - ${c.name}</option>`).join('')}
            </select>
          </div>
          <div class="form-group">
            <label class="form-label">Internal Assessment Score (Max 30)</label>
            <input type="number" class="form-control" name="internal" min="0" max="30" value="26" required /
>
          </div>
          <div class="form-group">
            <label class="form-label">Assignment Score (Max 20)</label>
            <input type="number" class="form-control" name="assignments" min="0" max="20" value="18" require
d />
          </div>
```

```
          <div class="form-group col-span-2">
            <label class="form-label">Semester Final Exam Score (Max 50)</label>
            <input type="number" class="form-control" name="exam" min="0" max="50" value="45" required />
          </div>
        </form>
      `;

      const footerHtml = `
        <button class="btn-quick" id="btn-cancel-grade">Cancel</button>
        <button class="btn-quick btn-primary" id="btn-save-grade">Calculate & Commit Grade</button>
      `;

      this.openModal('Faculty Grade Entry & Evaluation (IEEE 7384005)', bodyHtml, footerHtml);

      document.getElementById('btn-cancel-grade').addEventListener('click', () => this.closeModal());
      document.getElementById('btn-save-grade').addEventListener('click', () => {
        const form = document.getElementById('grade-entry-form');
        const fd = new FormData(form);

        const studentId = fd.get('studentId');
        const courseCode = fd.get('courseCode');
        const internal = parseInt(fd.get('internal'), 10) || 0;
        const assignments = parseInt(fd.get('assignments'), 10) || 0;
        const exam = parseInt(fd.get('exam'), 10) || 0;
        const total = internal + assignments + exam;
        const { grade, gpa } = this.bll.calculateGradeFromTotal(total);

        const courseObj = courses.find(c => c.code === courseCode);
        const student = this.dal.getStudentById(studentId);

        if (student) {
          student.scores = student.scores || [];
          const existingScoreIdx = student.scores.findIndex(sc => sc.courseCode === courseCode);
          const scorePayload = {
            courseCode,
            courseName: courseObj ? courseObj.name : courseCode,
            internal,
            assignments,
            exam,
            total,
            grade,
            gpa
          };

          if (existingScoreIdx !== -1) {
            student.scores[existingScoreIdx] = scorePayload;
          } else {
            student.scores.push(scorePayload);
          }

          // Recalculate CGPA
          student.cgpa = this.bll.computeStudentCGPA(student);
          this.dal.saveState();
          this.showToast(`Committed grade ${grade} (${gpa}) for ${student.name}`, 'success');
          this.closeModal();
          this.render();
        }
      });
    }

    openPostAnnouncementModal() {
      const bodyHtml = `
        <form id="announcement-form" class="form-grid">
          <div class="form-group col-span-2">
            <label class="form-label">Notice Title *</label>
```

```
            <input type="text" class="form-control" name="title" required placeholder="e.g. Schedule for Mid-Term Practical Exams" />
          </div>
          <div class="form-group">
            <label class="form-label">Publishing Authority *</label>
            <input type="text" class="form-control" name="author" value="Office of Academic Dean" required />
          </div>
          <div class="form-group">
            <label class="form-label">Priority Level</label>
            <select class="form-control" name="priority">
              <option value="Normal">Normal</option>
              <option value="Important">Important</option>
              <option value="High">High</option>
            </select>
          </div>
          <div class="form-group col-span-2">
            <label class="form-label">Notice Body *</label>
            <textarea class="form-control" name="content" required placeholder="Provide detailed institutional announcement text here..."></textarea>
          </div>
        </form>
      `;

      const footerHtml = `
        <button class="btn-quick" id="btn-cancel-ann">Cancel</button>
        <button class="btn-quick btn-primary" id="btn-post-ann">Publish Notice</button>
      `;

      this.openModal('Publish Campus Notice (IEEE 10516390)', bodyHtml, footerHtml);

      document.getElementById('btn-cancel-ann').addEventListener('click', () => this.closeModal());
      document.getElementById('btn-post-ann').addEventListener('click', () => {
        const form = document.getElementById('announcement-form');
        const fd = new FormData(form);

        const newAnn = {
          title: fd.get('title'),
          author: fd.get('author'),
          date: new Date().toISOString().split('T')[0],
          priority: fd.get('priority'),
          content: fd.get('content')
        };

        this.dal.addAnnouncement(newAnn);
        this.showToast('Published notice to all portal users!', 'success');
        this.closeModal();
        this.render();
      });
    }

    exportDatabaseJSON() {
      const dataStr = "data:text/json;charset=utf-8," + encodeURIComponent(JSON.stringify(this.dal.state, null, 2));
      const downloadAnchor = document.createElement('a');
      downloadAnchor.setAttribute("href", dataStr);
      downloadAnchor.setAttribute("download", `IMS_Database_Export_${new Date().toISOString().split('T')[0]}.json`);
      document.body.appendChild(downloadAnchor);
      downloadAnchor.click();
      downloadAnchor.remove();
      this.showToast('Exported central institutional database to JSON', 'success');
    }

    showToast(message, type = 'info') {
```

```
    let container = document.getElementById('toast-container');
    if (!container) {
      container = document.createElement('div');
      container.id = 'toast-container';
      container.className = 'toast-container';
      document.body.appendChild(container);
    }

    const toast = document.createElement('div');
    toast.className = `toast ${type}`;
    toast.innerHTML = `
      <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
        <circle cx="12" cy="12" r="10"></circle>
        <line x1="12" y1="8" x2="12" y2="12"></line>
        <line x1="12" y1="16" x2="12.01" y2="16"></line>
      </svg>
      <span>${message}</span>
    `;

    container.appendChild(toast);
    setTimeout(() => {
      toast.style.opacity = '0';
      toast.style.transform = 'translateY(10px)';
      toast.style.transition = 'all 0.3s ease';
      setTimeout(() => toast.remove(), 300);
    }, 3200);
  }
}

// Global bootstrap & Isomorphic Environment Detection
if (typeof window !== 'undefined' && typeof document !== 'undefined') {
  // Browser environment: initialize UI controller when DOM is ready
  document.addEventListener('DOMContentLoaded', () => {
    window.imsApp = new InstitutionManagementApp();
  });
} else if (typeof module !== 'undefined' && typeof process !== 'undefined') {
  // Node.js environment: auto-start server and open the browser
  try {
    const { startServer } = require('./server');
    console.log('Initiating IMS local server via app.js...');
    startServer({ openBrowser: true });
  } catch (err) {
    console.error('Failed to auto-start server from app.js:', err);
  }
}

```

#8 BAT start.bat

6 lines • 0.1 KB

Windows One-Click Startup Script

```
@echo off
title Educational Institution Management System (IMS) PoC
echo Starting IMS PoC Server...
node server.js
pause

```

#9 PLAINTEXT **.gitignore** 12 lines • 0.1 KB

Git Ignored Files Specification

```
node_modules/
npm-debug.log*
yarn-debug.log*
yarn-error.log*
.DS_Store
Thumbs.db
*.log
.env
.env.local
.vscode/
.idea/

```

#10 YAML .github/workflows/pages.yml

38 lines • 0.7 KB

GitHub Actions - Automated GitHub Pages Deployment

```yaml
name: Deploy Campus360 to GitHub Pages

on:
  push:
    branches:
      - main
      - Muhil
      - muhil
  workflow_dispatch:

permissions:
  contents: read
  pages: write
  id-token: write

concurrency:
  group: "pages"
  cancel-in-progress: false

jobs:
  deploy:
    environment:
      name: github-pages
      url: ${{ steps.deployment.outputs.page_url }}
    runs-on: ubuntu-latest
    steps:
      - name: Checkout
        uses: actions/checkout@v4
      - name: Setup Pages
        uses: actions/configure-pages@v5
      - name: Upload artifact
        uses: actions/upload-pages-artifact@v3
        with:
          path: '.'
      - name: Deploy to GitHub Pages
        id: deployment
        uses: actions/deploy-pages@v4

```

#11 JAVASCRIPT **build-code-pdf.js** 448 lines • 11.6 KB

Automated Full-Codebase PDF Documentation Generator

```
const fs = require('fs');
const path = require('path');
const { execSync } = require('child_process');

const projectDir = __dirname;
const outputFileHtml = path.join(projectDir, 'Campus360_Source_Code.html');
const outputFilePdf = path.join(projectDir, 'Campus360_Source_Code.pdf');

const filesToInclude = [
  { name: 'README.md', lang: 'markdown', desc: 'Project Overview, IEEE Research Mapping & System Architecture' },
  { name: 'package.json', lang: 'json', desc: 'Project Metadata, Dependencies & NPM Scripts' },
  { name: 'server.js', lang: 'javascript', desc: 'Zero-Dependency HTTP Server & Port Failover' },
  { name: 'index.html', lang: 'html', desc: 'Web Display Layer (Tier 1) - Multi-Role Dashboard HTML' },
  { name: 'styles.css', lang: 'css', desc: 'Styling & Visual Design System (CSS3 Variables & Responsive Layout)' },
  { name: 'data.js', lang: 'javascript', desc: 'Data Layer (Tier 4) - Seed Data & Entity Schemas' },
  { name: 'app.js', lang: 'javascript', desc: 'Business Logic Layer (Tier 2) & Data Access Layer (Tier 3)' },
  { name: 'start.bat', lang: 'bat', desc: 'Windows One-Click Startup Script' },
  { name: '.gitignore', lang: 'plaintext', desc: 'Git Ignored Files Specification' },
  { name: '.github/workflows/pages.yml', lang: 'yaml', desc: 'GitHub Actions - Automated GitHub Pages Deployment' },
  { name: 'build-code-pdf.js', lang: 'javascript', desc: 'Automated Full-Codebase PDF Documentation Generator' }
];

function escapeHtml(text) {
  return text
    .replace(/&/g, '&amp;')
    .replace(/</g, '&lt;')
    .replace(/>/g, '&gt;')
    .replace(/"/g, '&quot;')
    .replace(/'/g, '&#039;');
}

let tocItems = '';
let filesContentHtml = '';
let totalLines = 0;

filesToInclude.forEach((f, idx) => {
  const filePath = path.join(projectDir, f.name);
  if (!fs.existsSync(filePath)) return;

  const raw = fs.readFileSync(filePath, 'utf8');
  const lines = raw.split(/\r?\n/);
  totalLines += lines.length;

  tocItems += `
    <tr>
      <td class="toc-num">${idx + 1}</td>
      <td class="toc-file"><a href="#file-${idx}">${f.name}</a></td>
      <td class="toc-desc">${f.desc}</td>
      <td class="toc-lang">${f.lang.toUpperCase()}</td>
      <td class="toc-lines">${lines.length.toLocaleString()} lines</td>
    </tr>
  `;

```

```
    const numberedCode = lines
      .map((line, lIdx) => `<span class="line-num">${lIdx + 1}</span><span class="line-text">${escapeHtml(li
ne) || ' '}</span>`)
      .join('\n');

    filesContentHtml += `
      <div class="file-section" id="file-${idx}">
        <div class="file-header">
          <div class="file-meta">
            <span class="file-badge file-badge-num">#${idx + 1}</span>
            <span class="file-badge file-badge-lang">${f.lang.toUpperCase()}</span>
            <h2 class="file-title">${f.name}</h2>
          </div>
          <div class="file-stats">
            <span>${lines.length.toLocaleString()} lines</span>
            <span>&bull;</span>
            <span>${(raw.length / 1024).toFixed(1)} KB</span>
          </div>
        </div>
        <div class="file-description">${f.desc}</div>
        <pre class="code-block"><code>${numberedCode}</code></pre>
      </div>
    `;
  });

  const fullHtml = `<!DOCTYPE html>
  <html lang="en">
  <head>
  <meta charset="UTF-8">
  <title>Campus360 — Full Source Code Documentation</title>
  <style>
    @page {
      size: A4;
      margin: 14mm 12mm 14mm 12mm;
      @bottom-right {
        content: counter(page);
      }
    }

    * {
      box-sizing: border-box;
    }

    body {
      font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, "Helvetica Neue", Arial, sans-seri
f;
      color: #1e293b;
      background: #ffffff;
      margin: 0;
      padding: 0;
      font-size: 11pt;
      line-height: 1.5;
    }

    /* Cover Page */
    .cover-page {
      padding: 60px 30px;
      min-height: 90vh;
      display: flex;
      flex-direction: column;
      justify-content: space-between;
      page-break-after: always;
    }

    .cover-badge {
```

```
    display: inline-block;
    background: #eff6ff;
    color: #2563eb;
    border: 1px solid #bfdbfe;
    padding: 6px 14px;
    border-radius: 9999px;
    font-size: 10pt;
    font-weight: 600;
    letter-spacing: 0.05em;
    text-transform: uppercase;
    margin-bottom: 24px;
  }

  .cover-title {
    font-size: 30pt;
    font-weight: 800;
    color: #0f172a;
    line-height: 1.15;
    margin: 0 0 16px 0;
  }

  .cover-subtitle {
    font-size: 14pt;
    color: #475569;
    margin: 0 0 32px 0;
    line-height: 1.4;
  }

  .cover-card {
    background: #f8fafc;
    border: 1px solid #e2e8f0;
    border-radius: 8px;
    padding: 20px 24px;
    margin-bottom: 30px;
  }

  .cover-card h3 {
    margin: 0 0 12px 0;
    font-size: 12pt;
    color: #1e293b;
    font-weight: 700;
  }

  .cover-card ul {
    margin: 0;
    padding-left: 20px;
    color: #475569;
    font-size: 10pt;
    line-height: 1.6;
  }

  .cover-meta-grid {
    display: grid;
    grid-template-columns: repeat(2, 1fr);
    gap: 16px;
    margin-top: 20px;
  }

  .cover-meta-item {
    background: #f1f5f9;
    border-radius: 6px;
    padding: 12px 16px;
  }

  .cover-meta-label {
```

```
    font-size: 8pt;
    text-transform: uppercase;
    color: #64748b;
    font-weight: 600;
    letter-spacing: 0.05em;
  }

  .cover-meta-val {
    font-size: 11pt;
    font-weight: 600;
    color: #0f172a;
    margin-top: 4px;
    word-break: break-all;
  }

  /* Table of Contents */
  .toc-section {
    page-break-after: always;
    padding-top: 20px;
  }

  .toc-title {
    font-size: 20pt;
    font-weight: 800;
    color: #0f172a;
    margin-bottom: 8px;
  }

  .toc-subtitle {
    color: #64748b;
    font-size: 10pt;
    margin-bottom: 24px;
  }

  table.toc-table {
    width: 100%;
    border-collapse: collapse;
    font-size: 9.5pt;
  }

  table.toc-table th {
    text-align: left;
    padding: 10px 12px;
    background: #f1f5f9;
    border-bottom: 2px solid #cbd5e1;
    color: #334155;
    font-weight: 700;
    text-transform: uppercase;
    font-size: 8pt;
    letter-spacing: 0.05em;
  }

  table.toc-table td {
    padding: 10px 12px;
    border-bottom: 1px solid #e2e8f0;
    vertical-align: top;
  }

  .toc-num {
    font-weight: 700;
    color: #64748b;
    width: 30px;
  }

  .toc-file a {
```

```
    color: #2563eb;
    text-decoration: none;
    font-weight: 600;
    font-family: "Consolas", monospace;
  }

  .toc-desc {
    color: #475569;
  }

  .toc-lang {
    font-weight: 600;
    color: #64748b;
    font-size: 8pt;
  }

  .toc-lines {
    text-align: right;
    font-weight: 600;
    color: #334155;
    font-family: "Consolas", monospace;
  }

  /* File Sections */
  .file-section {
    page-break-before: always;
    margin-top: 10px;
  }

  .file-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    border-bottom: 2px solid #e2e8f0;
    padding-bottom: 10px;
    margin-bottom: 8px;
  }

  .file-meta {
    display: flex;
    align-items: center;
    gap: 10px;
  }

  .file-badge {
    padding: 4px 8px;
    border-radius: 4px;
    font-size: 7.5pt;
    font-weight: 700;
    text-transform: uppercase;
  }

  .file-badge-num {
    background: #0f172a;
    color: #ffffff;
  }

  .file-badge-lang {
    background: #dbeafe;
    color: #1e40af;
  }

  .file-title {
    margin: 0;
    font-size: 15pt;
```

```
      font-family: "Consolas", "Courier New", monospace;
      color: #0f172a;
    }

    .file-stats {
      font-size: 9pt;
      color: #64748b;
      display: flex;
      gap: 8px;
      font-family: "Consolas", monospace;
    }

    .file-description {
      font-size: 9.5pt;
      color: #475569;
      margin-bottom: 14px;
      font-style: italic;
    }

    /* Code Blocks */
    pre.code-block {
      margin: 0;
      padding: 12px;
      background: #f8fafc;
      border: 1px solid #e2e8f0;
      border-radius: 6px;
      font-family: "Consolas", "Courier New", monospace;
      font-size: 8pt;
      line-height: 1.45;
      color: #0f172a;
      white-space: pre-wrap;
      word-break: break-all;
      overflow: hidden;
    }

    .line-num {
      display: inline-block;
      width: 42px;
      color: #94a3b8;
      user-select: none;
      text-align: right;
      margin-right: 12px;
      border-right: 1px solid #e2e8f0;
      padding-right: 8px;
      font-size: 7.5pt;
    }

    .line-text {
      display: inline;
    }
  </style>
  </head>
  <body>

    <!-- Cover Page -->
    <div class="cover-page">
      <div>
        <div class="cover-badge">Source Code Documentation</div>
        <h1 class="cover-title">Campus360</h1>
        <div class="cover-subtitle">Educational Institution Management Information System (IMS) — Proof of C
oncept</div>

        <div class="cover-card">
          <h3>🏛 IEEE Research Paper Specifications Implemented</h3>
          <ul>
```

```
            <li><strong>IEEE 10058674</strong>: <em>"Institution Management System: Student Module"</em> — Record consolidation, dossier modal, promotion engine, placement & internship directorate.</li>
            <li><strong>IEEE 10516390</strong>: <em>"Educational Institution Management Information System"</em> — Centralized multi-role web platform (Admin, Faculty, Student) with institutional broadcast engine.</li>
            <li><strong>IEEE 7384005</strong>: <em>"College Student Management System Design Using Computer Aided System"</em> — 4-Tier Computer-Aided Architecture & Relational Entity-Relationship Model.</li>
          </ul>
        </div>
      </div>

      <div>
        <div class="cover-meta-grid">
          <div class="cover-meta-item">
            <div class="cover-meta-label">GitHub Repository</div>
            <div class="cover-meta-val">HaydenVictor420/Campus360</div>
          </div>
          <div class="cover-meta-item">
            <div class="cover-meta-label">Git Branch</div>
            <div class="cover-meta-val">muhil</div>
          </div>
          <div class="cover-meta-item">
            <div class="cover-meta-label">Local Server URL</div>
            <div class="cover-meta-val">http://localhost:3000</div>
          </div>
          <div class="cover-meta-item">
            <div class="cover-meta-label">Total Codebase Volume</div>
            <div class="cover-meta-val">${filesToInclude.length} files &bull; ${totalLines.toLocaleString()} total lines</div>
          </div>
        </div>
      </div>
    </div>

    <!-- Table of Contents -->
    <div class="toc-section">
      <h2 class="toc-title">Table of Contents</h2>
      <div class="toc-subtitle">Comprehensive listing of all project source code modules, schemas, and configurations.</div>
      <table class="toc-table">
        <thead>
          <tr>
            <th>#</th>
            <th>File Name</th>
            <th>Module Description</th>
            <th>Language</th>
            <th>Lines</th>
          </tr>
        </thead>
        <tbody>
          ${tocItems}
        </tbody>
      </table>
    </div>

    <!-- All Source Code Files -->
    ${filesContentHtml}

  </body>
  </html>
  `;

  fs.writeFileSync(outputFileHtml, fullHtml, 'utf8');
  console.log('Generated HTML preview at:', outputFileHtml);

  // Compile to PDF using Edge Headless
```

```
436  console.log('Compiling to PDF using Microsoft Edge...');
437  const edgePath = 'C:\\Program Files (x86)\\Microsoft\\Edge\\Application\\msedge.exe';
438  const cmd = `"${edgePath}" --headless=new --disable-gpu --no-pdf-header-footer --print-to-pdf="${outputFil
ePdf}" "${outputFileHtml}"`;
439
440  try {
441    execSync(cmd, { stdio: 'inherit' });
442    const stats = fs.statSync(outputFilePdf);
443    console.log(`Successfully generated PDF: ${outputFilePdf} (${(stats.size / 1024).toFixed(1)} KB)`);
444  } catch (err) {
445    console.error('Error generating PDF:', err.message);
446    process.exit(1);
447  }
448
```